

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics[†]

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Abstract

Extending saddle-path analysis to stochastic environments, I introduce *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by stochastic-equilibrium decision rules. I first use their *geometry* to diagnose state dependence: shock responses depend on position along the saddle and on slope gaps across regimes. I then show that their *monotonicity* delivers dimension reduction within a frozen-regime spell, since aggregate capital exactly indexes the equilibrium state along each conditional saddle in [Krusell and Smith \(1998\)](#). Across spells, a synchronization theorem bounds the residual gap between histories by the regimes they recently shared. Finally, extending the geometry to a two-dimensional conditional saddle *surface* with an investment externality, I analyze the optimal investment tax credit surface over the state space.

Keywords: Conditional saddle path, business cycles, state-dependent dynamics, sufficient statistics, heterogeneous agents, investment externality.

JEL codes: C62, D58, E32.

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1 Introduction

Understanding equilibrium dynamics in models with aggregate uncertainty remains a central challenge in macroeconomics. Unlike deterministic models, where saddle-path diagrams provide immediate geometric intuition, stochastic equilibrium models lack comparable geometric frameworks. This makes it difficult to develop intuition about how economies respond to shocks, how different states interact, and why certain computational methods work. The challenge becomes particularly acute in heterogeneous-agent models, where the natural state variable is an infinite-dimensional wealth distribution, yet computational work routinely achieves dimension reduction to low-dimensional aggregates. Can we extend geometric saddle-path analysis to stochastic environments? What does such a framework reveal about equilibrium dynamics, state-dependent responses, and the success of computational approximations?

The paper makes three contributions, all drawn from a single geometric object. First, I define *conditional saddle paths*: equilibrium trajectories under frozen exogenous regimes, traced by the decision rules of the stochastic equilibrium. Each regime’s trajectory draws a policy graph in the aggregate phase plane, decomposing fluctuations into movements *along* a conditional saddle (endogenous propagation) and *across* saddles (exogenous shocks), the stochastic analogue of the deterministic phase diagram. The geometry reveals all the possible dynamics within the equilibrium, both factual and counterfactual, without separate numerical results: the model’s entire dynamics are visible in a single picture. A stark example is the shock response and its state dependence: impacts are state-independent when the saddles are parallel; relative tilt makes the same shock hit harder in some states; and even with parallel saddles, the *shape* of the transition (peak timing and persistence) depends on position along the saddle. The graphs also visualize equilibrium-level comparative statics: a changed parameter (e.g., risk aversion) re-solves the recursive competitive equilibrium and tilts the saddles, showing how the parameter reshapes the equilibrium dynamics. Set against their perfect-foresight counterparts, the saddles yield a *flow anticipation wedge*: the area between the two families is the consumption (dis-)saved ahead of regime change, surplus geometry on the state space. Its welfare counterpart measures the value of anticipated reversion in consumption equivalents, visualizing the net surplus from the business cycle in the spirit of [Lucas \(1987\)](#). I develop these in a canonical RBC model (Sections 2 and 3).

Second, the same geometry speaks to a long-standing computational puzzle: why tracking a single aggregate can stand in for an entire wealth distribution. When an aggregate

variable moves monotonically along the conditional saddle, it exactly indexes the equilibrium state within a frozen-regime spell, so one number recovers the distribution and every price. I show that aggregate capital plays this role in [Krusell and Smith \(1998\)](#): within each spell the dimension reduction is exact along the entire conditional saddle. The key is the invariance of the phase diagram to heterogeneity: the heterogeneous economy's conditional K -nullclines coincide with those of the canonical RBC economy, because the zero-net-investment boundary is an accounting identity in aggregates, independent of *who* saves. This exact sufficiency is within-spell only: two histories can reach the same K carrying different distributions. Across spells, a synchronization theorem bounds this residual gap by the regimes the two histories recently shared. The contraction rate is anchored near the conditional steady state; the remaining discrepancy, small in [Krusell and Smith \(1998\)](#), is documented directly. Together they yield the near-global sufficiency of aggregate capital, documented directly in this calibration. The underlying monotonicity is verified two ways: a cheap order-preservation check on the computed frozen-regime map holds on every realized spell of this economy, and bounds on MPCs and price feedback, checked along the computed continuation, deliver it on an explicit region (Section 4.1).

Third, I extend the geometry to two predetermined states, where the conditional saddle becomes a *surface*. In an RBC economy where investment builds knowledge that firms do not pay for (Section 4.2), the planner traces a second surface, resting at higher capital and knowledge, and a state-contingent investment tax credit implements it. That credit is itself a surface over the two stocks, so optimal policy can be analyzed geometrically on the same state space. The surface is countercyclical and tilted, rising in capital and falling in knowledge; matching the planner's steady state is therefore weaker than implementing its policy along transitions.

Related literature This paper contributes to three strands of literature in macroeconomics. The first studies global equilibrium dynamics and solution methods under aggregate uncertainty, where characterizing equilibrium dynamics with aggregate shocks has motivated extensive methodological development. [Marcet \(1988\)](#) and [Den Haan and Marcet \(1990\)](#) pioneered simulation-based solution methods that parameterize the expectation function under rational expectations; [Krusell and Smith \(1998\)](#) instead posited boundedly rational forecasters and found that a simple linear forecasting rule in aggregate capital achieves remarkable accuracy ($R^2 > 0.9999$) despite the infinite-dimensional state space. Subsequent work sharpened accuracy and efficiency through perturbation and projection methods ([Re-](#)

iter, 2010; Winberry, 2018; Bhandari et al., 2023), simulation and aggregation approaches (Den Haan and Rendahl, 2010; Judd et al., 2011), sequence-space methods (Auclert et al., 2021; Azinovic-Yang and Žemlička, 2025), and deep learning (Maliar et al., 2021; Azinovic et al., 2022; Fernández-Villaverde et al., 2023; Han et al., 2026). Childers (2018) instead linearizes in function space, keeping the whole distribution as the state. On the theoretical side, Miao (2006) and Cao (2020) establish existence of competitive and recursive equilibria in this class; Proehl (2025b) gives conditions for existence and uniqueness under aggregate and idiosyncratic risk, and Proehl (2025a) develops a companion global solution method. The framework here is complementary, taking the recursive equilibrium as given and characterizing the geometry of its dynamics.

The conditional steady state links this framework to Markov-switching DSGE, where regime-dependent steady states anchor decision rules and local perturbations (Chung et al., 2007; Bianchi, 2016; Foerster et al., 2016).¹ Building on these, I study the entire path converging to such a point, capturing the dynamics globally rather than only in its neighborhood: the path yields the whole equilibrium dynamics nested within the interval the conditional steady states bracket, an ordering of those states against their perfect-foresight counterparts (Proposition 4), a monotone coordinate for the distributional state (Theorem 1), and the slope and curvature carrying state dependence.

The sequence-space Jacobian method of Auclert et al. (2021) reduces dimension by linearizing around a steady state; conditional saddle paths instead operate globally, exploiting monotonicity of an aggregate coordinate along the frozen-regime path. A first-order method reproduces the saddle’s tangent at its anchor, the deterministic steady state, which Proposition 4 places away from either K_A^{cs} ; elsewhere the saddle bends, capturing state-dependent dynamics that first-order methods cannot, such as the regime-dependent slope asymmetries frictions generate (Section 3).

This paper differs from these computational contributions by providing geometric foundations for *why* dimension reduction works: aggregate capital’s within-spell sufficiency follows from nullcline invariance and monotonicity rather than numerical coincidence. This also has computational implications, since along each conditional saddle aggregate capital provides a theoretically justified metric for the repeated transition method of Lee (2026a).²

¹Each K_A^{cs} is stationary under its own regime alone, and its failure to be stationary after a switch is what carries the economy across saddles.

²Freezing the aggregate state and analyzing transitions appears in several computational approaches to heterogeneous-agent models, including Bourany (2018), Section 4.3 of a working paper version of Achdou et al. (2022), and Lee (2026a). In continuous time, the master-equation literature (Achdou et al., 2022; Bilal,

Second, this paper extends the geometric tradition of phase-diagram analyses in growth theory (Solow, 1956; Ramsey, 1928; Koopmans, 1963; Cass, 1965) to stochastic environments. Relatedly, the stable transformation path of Buera et al. (2024) extends the balanced growth path, and its turnpike property, to non-stationary growth models, defining a stable path independent of initial conditions. Conditional saddle paths extend the same classical geometry along a different dimension, from deterministic to stochastic settings, providing a visual tool for how microfounded frictions generate state-dependent dynamics (Kaplan and Violante, 2014; Vavra, 2014; Berger and Vavra, 2015; Basu and Bundick, 2017; Bloom et al., 2018; Kaplan et al., 2018; Petrosky-Nadeau et al., 2018; Rognlie et al., 2018; Baley and Blanco, 2019; Pizzinelli et al., 2020; Berger et al., 2021; Winberry, 2021; Lee, 2026b; Melcangi, 2024).

Finally, the random dynamical systems literature (Arnold, 1998; Schenk-Hoppé, 2001) focuses on history-dependent stochastic saddle paths (Yannacopoulos, 2011) that vary with the realized shock sequence.³ By contrast, conditional saddle paths are history-invariant geometric objects, the direct analogue of deterministic phase diagrams.

2 Conditional saddle path

2.1 Definitions and assumptions

I consider a generic dynamic stochastic model whose recursive competitive equilibrium (RCE) is characterized by an aggregate state S and laws of motion $(\Gamma_{\text{endo}}, \Gamma_{\text{exo}})$:

$$S = [\Phi, A] \tag{1}$$

where Φ is the endogenous and A the exogenous aggregate state, the latter possibly multivariate. The exogenous law Γ_{exo} is a Markov chain; for the illustration I take a two-state chain with $A \in \{B, G\}$ and $\Gamma_{\text{exo}}(A'|A) > 0 \forall A', A$.

The distributional state space $\mathcal{X}$ has elements Φ summarizing the cross-sectional distribution of payoff-relevant idiosyncratic household states (e.g., assets and productivity types)

(2023) treats the distribution itself as the state and asks when finite-dimensional projections suffice, the rigorous mathematical home of this question. Conditional saddle paths formalize the underlying geometric objects, and the sufficiency results explain when dimension reduction is exact along frozen-regime paths.

³In Random Dynamical Systems language, the conditional steady state is a deterministic realization of the random fixed point (Schenk-Hoppé and Schmalfuß, 2001), and the conditional saddle path is the invariant manifold of the frozen-regime dynamics. The synchronization theorem of Section 4.1 (Theorem 2) uses exactly this pullback random-fixed-point construction, recovering the current state as the long-history limit of trajectories started ever further in the past under the same regime history.

plus any endogenous objects needed to evaluate decision rules. Formally, $\mathcal{X}$ is a subset of a metric space of probability measures, possibly augmented by a finite-dimensional vector of aggregate variables. Its metric d treats two states as close when the averages they assign to any well-behaved function of the individual state are close, with the finite-dimensional components compared in the Euclidean norm.⁴ Throughout, $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ maps the current distributional state to next period's under frozen regime A .

The central object formalizes a natural thought experiment: the economy sits at (Φ_0, A) today and the regime stays at A indefinitely. The resulting equilibrium trajectory, computed under decision rules that fully internalize future regime changes, is a frozen-regime orbit. Its closure (visited states plus their limit points) is the conditional saddle path. In aggregate phase diagrams the associated regime-specific locus is a conditional policy graph, to which the slope and curvature statements below refer.

Definition 1 (Frozen-regime orbit and conditional saddle path).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ denote the frozen-regime continuation under A , defined recursively by

$$\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A), \quad t \geq 0. \quad (2)$$

The frozen-regime orbit is

$$O(\Phi_0, A) := \{\Phi_t : t \geq 0\}. \quad (3)$$

The conditional saddle path under A from Φ_0 is defined as the closure of the frozen-regime orbit:

$$\mathcal{M}(\Phi_0, A) := \overline{O(\Phi_0, A)}. \quad (4)$$

Two projection components recur: a *coordinate* $e : \mathcal{X} \rightarrow \mathbb{R}$, a predetermined aggregate variable such as capital K or a knowledge stock H ; and an *aggregate object* $g(\Phi, A)$, an equilibrium outcome such as consumption or investment.

Definition 2 (Conditional policy graph).

For a regime A , a coordinate e , and an aggregate object g , the conditional policy graph in the (e, g) projection is

$$\mathcal{P}_A^g := \{(e(\Phi), g(\Phi, A)) : \Phi \in \mathcal{D}_A\}, \quad (5)$$

⁴ d is any metric for which $d(\Phi_n, \Phi) \rightarrow 0$ if and only if $\int f d\Phi_n \rightarrow \int f d\Phi$ for every bounded continuous function f ; this is the standard notion of weak convergence.

where $\mathcal{D}_A := \bigcup_{\Phi_0 \in \mathcal{X}_0} \mathcal{M}(\Phi_0, A)$ for a specified set of initial states $\mathcal{X}_0 \subseteq \mathcal{X}$, suppressed in the notation; the applications take $\mathcal{X}_0$ to be the simulated ergodic support or, in Section 4.2, the sampled basin.

When e is injective (one-to-one) on a given conditional saddle path, the restriction of $\mathcal{P}_A^g$ to that path is the graph of a single-valued function of e . Over the pooled domain $\mathcal{D}_A$ the graph is in general a correspondence: distinct orbits can pass through the same e at different g (Section 4.1). When $e(\Phi)$ is itself the full endogenous state, as in the representative-agent economies below, g depends on Φ only through e and the pooled graph is single-valued.

Based on the conditional saddle path, I define the conditional steady state:

Definition 3 (Conditional steady state).

Given (Φ_0, A) , if the frozen-regime continuation $\{\Phi_t\}_{t \geq 0}$ converges, I denote its limit by

$$\Phi^{cs}(\Phi_0, A) := \lim_{t \rightarrow \infty} \Phi_t. \quad (6)$$

Write $K_A^{cs} := K(\Phi^{cs}(\Phi_0, A))$, suppressing Φ_0 when context fixes it.

A *spell* is a maximal run of dates with A constant, along which the equilibrium path is the initial segment of this continuation. Fixing A makes the law of motion deterministic on the distribution space: from Φ_0 , $\{\Phi_t\}_{t \geq 0}$ is generated by repeated application of $\Gamma_{\text{endo}}(\cdot, A)$. Thus $\mathcal{M}(\Phi_0, A)$ is the analogue of the saddle arm in deterministic saddle-path analysis, the equilibrium states visited under regime A together with their limit points.⁵

For the conditional saddle to serve as a useful geometric object, a one-dimensional curve traversed during each frozen-regime spell, the dynamics must converge to a unique limit and vary smoothly. The following assumption collects these conditions.

Assumption 1 (Regularity of conditional saddle paths).

Fix a regime $A \in \{B, G\}$ and an initial state $\Phi_0 \in \mathcal{X}$. Let $\{\Phi_t\}_{t \geq 0}$ be defined by $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$.

- (i) (Unique existence) On $\mathcal{M}(\Phi_0, A)$, the conditional steady state $\Phi^{cs}(\Phi_0, A)$ uniquely exists.
- (ii) (Continuity) The map $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is continuous. Moreover, the aggregate variables (e.g., aggregate capital K and consumption C) vary continuously in Φ when restricted to $\mathcal{M}(\Phi_0, A)$.

⁵Strictly, in discrete time $\mathcal{M}(\Phi_0, A) = \overline{\mathcal{O}(\Phi_0, A)}$ is an orbit closure rather than a smooth manifold, and the saddle-path label refers to its path-like role. Slope and curvature therefore refer to the policy graph $\mathcal{P}_A^g$. In continuous time the frozen-regime object is a one-dimensional invariant manifold.

Part (i) requires that, conditional on regime A and initial state Φ_0 , the frozen-regime continuation converge to a unique conditional steady state; it asserts neither global uniqueness across initial conditions nor the absence of multiplicity across regimes.⁶ Part (ii) imposes continuity of the law of motion and aggregate observables on the conditional saddle, so small perturbations of the state produce small changes in aggregate outcomes. Assumption 1(i) thus restricts attention to convergent continuations, ruling out periodic orbits or divergent paths. In the RBC benchmark convergence follows from diminishing returns: the marginal product of capital is high below the conditional steady state and low above it, so capital approaches it from either side. In the heterogeneous-household economy convergence is instead a maintained condition on the equilibrium law of motion.

With this regularity the conditional saddle carries a notion of position along the orbit. Theorem 1 (Section 4.1) makes the implication exact: an aggregate variable that moves strictly one-way along the path is a coordinate for the entire conditional saddle, an exact one-dimensional representation on $\mathcal{M}(\Phi_0, A)$. The result is *pathwise*, orbit by orbit, regardless of the number of predetermined states; with two or more stocks the orbits fill a *surface* rather than a curve, on which a single monotone coordinate need not exist. Section 4.2 analyzes this multi-state case, where the surface also holds the paper’s planner–equilibrium geometry.

2.2 Representative-agent economy: A canonical RBC

A representative household with log utility maximizes lifetime utility under stochastic aggregate TFP A_t , given (a_0, A_0) . I use the recursive competitive equilibrium, equivalent to the sequential equilibrium under standard regularity conditions. The recursive form is:

$$v(a; K, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a'; K', A') \quad (7)$$

$$c + a' = a(1 + r(K, A)) + w(K, A) \quad (8)$$

$$a' \geq -\bar{a} \quad (9)$$

$$K' = \Gamma_{\text{endo}}(K, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A), \quad (10)$$

where v is the value function, a beginning-of-period wealth, K aggregate capital, A aggregate TFP, and Γ_{endo} the law of motion for K . TFP A follows a Markov-switching process between

⁶Proehl (2025b) establishes global existence and uniqueness of the recursive equilibrium under explicit conditions on primitives, whereas Assumption 1(i) allows global multiplicity as in Walsh and Young (2024); conditional on a given initial state and fixed regime, the equilibrium path is assumed uniquely determined.

levels B and G , with $G > B$:

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix} \quad (11)$$

Competitive factor prices from a CRS Cobb–Douglas production function are:

$$\begin{aligned} r(K, A) &= A\alpha K^{\alpha-1} - \delta, \\ w(K, A) &= A(1 - \alpha)K^{\alpha}, \end{aligned} \quad (12)$$

where capital K satisfies $K = a$ in equilibrium. The recursive competitive equilibrium (RCE) is:

Definition 4 (Recursive competitive equilibrium).

$(c, a', v, r, w, \Gamma_{\text{endo}})$ is a recursive competitive equilibrium if:

1. given prices (r, w) , the policies (c, a') attain the value v in the household problem;
2. prices satisfy the competitive factor-price conditions (12);
3. consistency holds:

$$a'(K; K, A) = K' = \Gamma_{\text{endo}}(K, A). \quad (13)$$

Using Γ_{endo} from Definition 4, I define the conditional K -nullcline.

Definition 5 (Conditional K -nullcline in RBC).

Fix a regime $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the (K, C) phase space,

$$\mathcal{N}(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (14)$$

where $\Delta K(K, C; A)$ is the one-period change in capital implied by the resource constraint at (K, C) under regime A . When $\mathcal{N}(A)$ is the graph of a function, I write $\mathcal{N}(A) = \{(K, C) : C = C_A^{Knull}(K)\}$.

Along a strictly monotone frozen-regime continuation, the orbit meets $\mathcal{N}(A)$ only in the limit, at the conditional steady state; the locus is thus distinct from the *realized nullcline crossings* $\{(K_t, C_t) : t \geq 0, K_{t+1} = K_t\}$, empty before the limit. The resource constraint $Y = C + I$ with $I = K_{t+1} - (1 - \delta)K_t$ gives $\Delta K = AK^{\alpha} - \delta K - C$, so $\mathcal{N}(A)$ is an explicit function graph.

Proposition 1 (Characterizing the conditional K -nullcline in RBC).

The conditional nullclines of aggregate capital K for $A \in \{B, G\}$ are

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (15)$$

independent of any initial condition.

Proof. From $\Delta K(K, C; A) = 0$ and $\Delta K = AK^\alpha - \delta K - C$, the nullcline is $C = AK^\alpha - \delta K$ for each A . ■

The nullcline invokes only $\Delta K = AK^\alpha - \delta K - C$, so it bears no imprint of preferences, beliefs, or the distribution; the same holds for the induced *direction field*, whose net-investment sign $\Delta K = C_A^{Knull}(K) - C$ is fixed by accounting alone. Beliefs and heterogeneity relocate the equilibrium orbit within a phase portrait whose boundary and sign structure they cannot move. This is the common root of belief-invariance under aggregate uncertainty (Proposition 3), underlying the monotonicity diagnostic of Appendix G,⁷ and distribution-invariance in the heterogeneous-household model (Proposition 7), underlying within-spell dimension reduction.

In discrete time capital can jump a finite amount, so the projected (K, C) trajectory can develop folds or loops, pathologies absent in continuous time. I rule them out with a geometric restriction.

Assumption 2 (No segment crossing of the conditional K -nullcline).

For each $A \in \{B, G\}$, along the frozen- A continuation: for every pre-limit transition (every $t \geq 0$ with $K_{t+1} \neq K_t$), the open straight line segment in (K, C) connecting (K_t, C_t) and (K_{t+1}, C_{t+1}) does not intersect the active regime's conditional K -nullcline $\mathcal{N}(A)$:

$$((K_t, C_t), (K_{t+1}, C_{t+1})) \cap \mathcal{N}(A) = \emptyset, \quad \forall t \geq 0 \text{ with } K_{t+1} \neq K_t. \quad (16)$$

Endpoints are not required to lie off the nullcline; in particular, the conditional steady state is allowed to lie on $\mathcal{N}(A)$, and the assumption imposes no condition at the degenerate terminal transition $K_{t+1} = K_t$.

Assumption 2 ensures consecutive points cannot strictly switch sides of the K -nullcline. Its companion is *unique K -stationarity*: $K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$, which in the representative-agent model follows from Assumption 1(i) (K is Markov under the frozen regime) and is otherwise maintained as a pathwise hypothesis. Economically, Assumption 2

⁷All lettered appendices are collected in the Online Appendix.

rules out single-period overshooting so severe that capital jumps past its stationary level and reverses, the discrete-time counterpart of the orbit meeting $\mathcal{N}(A)$ only in the limit.⁸

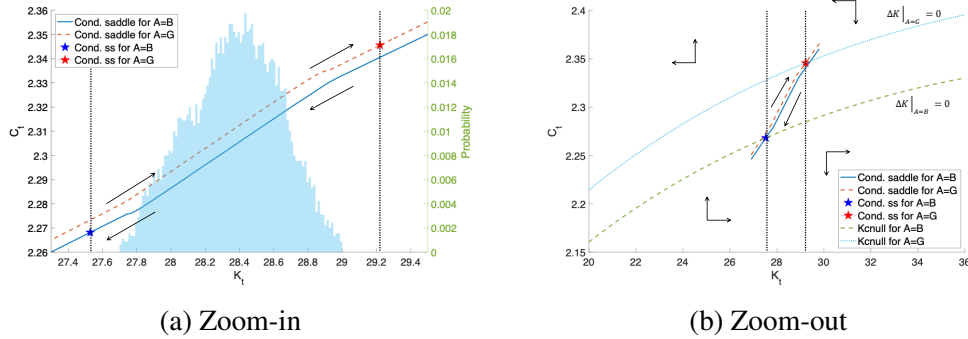
Although stated for the RBC benchmark, Assumption 2 is central to the heterogeneous-household economy (Section 4.1). It can be verified from primitives without solving the full stochastic equilibrium, conditional on a simulated frozen-regime continuation (or certified envelope bounds along it). Appendices G and E develop two complementary strategies.

Proposition 2 (*K-monotonicity in RBC*).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique *K*-stationarity ($K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$ along the frozen-*A* continuation), aggregate capital converges to K_A^{cs} strictly monotonically.

Proof. Fix A and let $D_t := C_t - C_A^{Knull}(K_t)$ denote the nullcline gap. The resource constraint gives $\Delta K_t = -D_t$. By Assumption 2, no transition strictly switches sides of the nullcline: $D_t D_{t+1} \geq 0$ for every transition t with $K_{t+1} \neq K_t$, and $D_t \neq 0$ on any such transition. Unique *K*-stationarity confines $D_{t+1} = 0$ to terminal arrival at Φ_A^{cs} , so prior to convergence $\{\Delta K_t\}$ has constant nonzero sign. Unique *K*-stationarity rules out nonterminal flat steps ($\Delta K_t = 0$ at any $\Phi_t \neq \Phi_A^{cs}$), so every pre-convergence transition is nondegenerate and $\{K_t\}$ is strictly monotone. Convergence $K_t \rightarrow K_A^{cs}$ follows from Assumption 1. ■

Figure 1: Conditional saddle paths and the phase diagram



Notes: Conditional *K*-nullclines ($\Delta K = 0$) partition the (K, C) plane. Direction arrows show frozen-regime convergence along each conditional saddle. Panel (a) zooms into the neighborhood of the conditional steady states shown in panel (b). The histogram plots the time-series distribution of aggregate capital.

Unique *K*-stationarity as a no-plateau condition Unique *K*-stationarity rules out *plateau equilibria*: states off the conditional steady state at which aggregate capital stops moving

⁸For related geometric restrictions in dynamic optimization, see Reddy et al. (2020) and, on threshold (Skiba) phenomena, Skiba (1978), Dechert and Nishimura (1983), and Wagener (2003).

($\Delta K_t = 0$) while the distribution keeps evolving, excluding the degenerate fixed points of K that arise with poverty traps, occasionally binding constraints, or policy kinks. In the representative-agent RBC, K is Markov under the frozen regime, so the condition reduces to uniqueness of the conditional steady state (Assumption 1(i)); in the heterogeneous-agent economy it holds locally if the linearized frozen-regime dynamics admit a stable eigenvalue $\lambda_A^s \in (0, 1)$, which Proposition 10 assumes rather than derives. A globally certified version is more delicate: the bounded price-income MPC of Assumption 3 (Appendix E) constrains the off-steady-state dynamics, but I treat unique K -stationarity as a maintained pathwise condition, checkable along any simulated frozen-regime continuation.

In the representative-agent benchmark these restrictions are not free-standing assumptions but consequences of the scalar concave structure, given two checkable inputs verified in the computed equilibrium: a curvature-margin bound and, for stochastic beliefs, a single-crossing condition on the conditional Euler residual. Since capital is the only state, the frozen- A continuation iterates a single policy $g_A(K)$, and Appendix J derives each restriction from a primitive: strict concavity delivers the unique conditional steady state, strictly monotone convergence, and, via the curvature-margin bound, no segment crossing, a bound standard calibrations satisfy by orders of magnitude; feasibility bounds capital; transversality enters only in selecting the stable policy. The conditions are stated as model assumptions because in the heterogeneous-agent economy of Section 4.1, the iterated state is the whole distribution, not scalar capital. Next period's K then depends on the distribution rather than on K alone. Monotone convergence requires a real stable eigenvalue of the linearized distributional dynamics, and primitives do not pin one down.

Figure 1 depicts the calibrated RBC model's conditional saddle paths in the (K, C) plane under $A \in \{B, G\}$, solved globally via the repeated transition method of Lee (2026a).⁹ Panel (a) zooms into the conditional steady states with a histogram of simulated K_t ; panel (b) shows the global partition with conditional K -nullclines and direction arrows. *All* the possible dynamics of the equilibrium run along the two conditional saddles, over the capital interval bounded by their conditional steady states: within a spell the economy converges along the active saddle, and regime switches move it across; every equilibrium path, factual or counterfactual, is an alternation of these two motions, so the entire ergodic distribution is nested inside this interval.

⁹The baseline is quarterly: $\beta = 0.99$, $\alpha = 0.33$, $\delta = 0.025$, $A \in \{0.99, 1.01\}$ with $\pi_{AA} = 0.90$ (Appendix A). For each regime A , I hold productivity fixed for 2,500 periods and iterate the equilibrium law of motion from initial capital on both sides of K_A^{cs} , recovering the full conditional saddle in the phase diagram.

MIT shock on the saddle This decomposition characterizes unexpected transitory (MIT) shocks, whose link to first-order perturbation in heterogeneous-agent models is established by [Boppart et al. \(2018\)](#). For shocks representable as displacements of the predetermined state under unchanged decision rules, a shift along the conditional saddle replicates the impact, and all subsequent dynamics are along-path convergence. The slope and curvature of the conditional policy graph at the displacement point thus determine propagation, including persistence, amplification, and state dependence.

Generalized transition function Because conditional saddle paths capture all equilibrium transitions, along and across saddles, they support generalized transition function analysis ([Lee, 2026a](#)), encompassing generalized impulse response functions ([Koop et al., 1996](#); [Andreassen et al., 2018](#)). Figure 2 shows equilibrium dynamics in the phase diagram (panel (a)) and time domain (panel (b)) over a 20-quarter subsample. Negative TFP shocks produce downward jumps in consumption across conditional saddles, followed by endogenous decline along them; positive shocks reverse this. No step is confined to local dynamics around a steady state, and the phase diagram holds the whole path in one figure. For the GIRF,

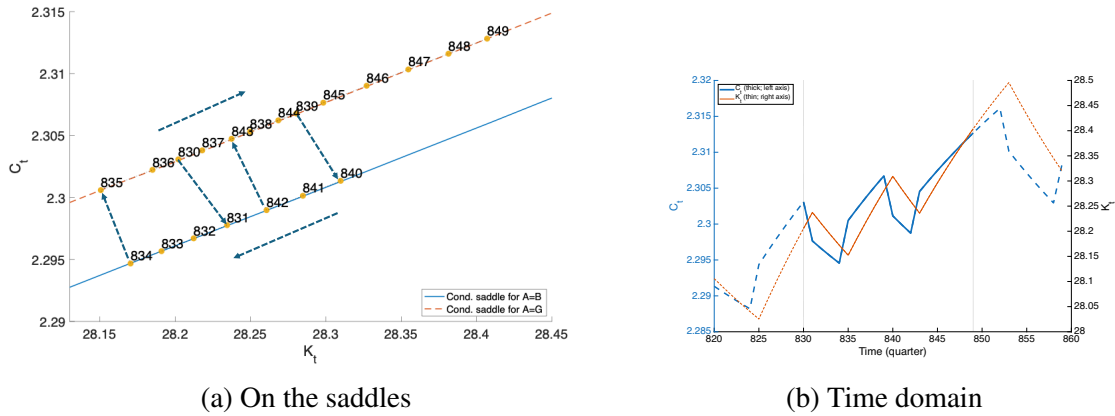


Figure 2: Equilibrium dynamics: saddle vs. time domain

Notes: Stochastic equilibrium dynamics on the conditional saddle paths (panel (a)) and in the time domain (panel (b), C_t thick on the left axis, K_t thin on the right; dashed segments lie outside the window). A 20-quarter window from the simulated path (quarters 830–849).

one can replace the two-state chain with a finely discretized AR(1) for log productivity, $\ln A' = \rho \ln A + \epsilon$, $\epsilon \sim N(0, \sigma^2)$. There is then a continuum of frozen- A conditional saddles indexed by A , and the GIRF path moves both along a given saddle (endogenous propagation) and across the family as A evolves.

Following a positive TFP shock, consumption jumps upward across conditional sad-

dles on impact; thereafter, along- and across-saddle forces jointly generate a bow-shaped trajectory in the phase diagram. Capital is hump-shaped as the shock mean-reverts, and consumption first rises then falls as TFP crosses the level at which along-saddle adjustment reverses direction.

Comparison with the perfect-foresight saddles In the *perfect-foresight* (PF) economy, agents believe the current regime persists forever, so the PF steady states solve $1 = \beta(1 - \delta + \alpha A(K_A^{pf})^{\alpha-1})$. Along the *conditional saddle* (CS), the regime is fixed ex post but agents correctly anticipate switches under Γ_{exo} , so the conditional steady states K_A^{cs} are pinned by probability-weighted Euler equations.¹⁰ The PF and CS economies share the same K -nullcline (Proposition 3) but differ in consumption dynamics.

Proposition 3 (K -nullcline invariance over beliefs).

Conditional K -nullclines are identical between the RBC model with aggregate uncertainty and the perfect-foresight counterpart.

Proof.

From the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$, equation (15) is immediate. Neither condition involves expectations or belief structures, so the nullcline is identical whether or not there is aggregate uncertainty. ■

PF–CS differences thus come entirely from consumption dynamics, not the feasibility locus; Proposition 4 formalizes this via steady-state orderings. This invariance is an accounting identity independent of beliefs and, in Section 4.1, of *who* saves, on which both the dimension-reduction result and the monotonicity diagnostic rest.

Proposition 4 (Aggregate uncertainty and the conditional steady states).

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return and $c(K, A')$ the stochastic-equilibrium consumption function. Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (17)$$

and (ii) each conditional Euler residual F_A , the residual of the conditional steady-state Euler equation viewed as a function of K (Appendix A), is strictly positive below K_A^{cs} and

¹⁰Specifically, $\beta\pi_{AA}(1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta) + \beta\pi_{AA'}(c(K_A^{cs}, A')/c_A^{cs})^{-1}(1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta) = 1$ for each $A \in \{B, G\}$, where $A' \neq A$.

strictly negative above it. Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (18)$$

Proof.

See Appendix A. ■

Both hypotheses are explicit and directly checkable. Upon a persistent regime switch, consumption moves roughly one-for-one while the gross return moves only through its marginal-product term, so jump dominance (17) holds with a wide margin here, 0.77 percent against a 0.07 percent threshold (Appendix A). Under full depreciation ($\delta = 1$) the decision rules collapse to the belief-independent form of Brock and Mirman (1972), (17) holds with equality, and the two sets of steady states coincide (Appendix A); away from that corner the orderings (18) hold in the computed equilibrium.

The two families are nested in opposite directions. The steady states nest one way: the PF pair lies inside the CS pair. Anticipated regime reversion tilts each conditional Euler equation at first order, and households (dis-)save ahead of the switch in the spirit of buffer-stock behavior (Deaton, 1991; Carroll, 1997), pushing the conditional steady states apart. The *paths* nest the other way: at any common K , pricing the switch pulls each regime's consumption toward the cross-regime average, so the CS saddles run between the PF saddles. The two nestings are one mechanism: compressing consumption toward the average is (dis-)saving, and accumulating it carries the resting points apart.

Panel (a) of Figure 3 shades the path gap: one-signed regime by regime, and equal to the net-investment wedge state by state.

Panel (a) reads like the textbook surplus diagram, with the state in place of quantity: the vertical distance between the two curves is a consumption flow, and the area accumulates it across states. The B region is a consumption surplus against the perfect-foresight benchmark, financed one-for-one by slower accumulation; the G region is the consumption loss, matched by faster accumulation. Panel (b) prices the process. A conditional value saddle plots the household's value along the frozen- A orbit, gaps converted at $\exp[(1 - \beta)\Delta V] - 1$. Each runs parallel to its perfect-foresight counterpart, a one-signed gap of +1.26 percent of consumption under B and -1.23 under G . In consumption-equivalent units (Lucas, 1987), this is the state-conditional welfare value of anticipated regime reversion relative to regime permanence: positive where reversion promises recovery, negative where it threatens relapse.¹¹ The

¹¹The absorbing benchmark bundles the direct effect of future switches with the saving response they induce. Evaluating the CS and PF policies under the same switching law isolates the welfare gain from anticipatory

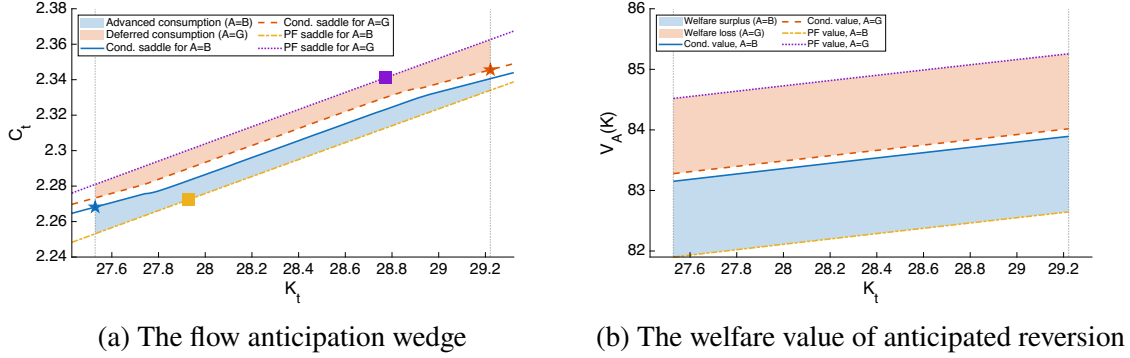


Figure 3: Policy and welfare effects of anticipated regime reversion

Notes: Panel (a) shades $C_A^{cs}(K) - C_A^{pf}(K)$ on $[K_B^{cs}, K_G^{cs}]$ (dotted verticals), +0.47 and -0.48 percent of consumption at the PF steady states; stars mark conditional and squares PF steady states. Panel (b): value saddles under the switching law against the absorbing (regime-permanence) benchmark; the gap is +1.26 and -1.23 percent in consumption equivalents.

near-cancellation across the two shaded gaps is geometric: symmetric primitives make their stationary-weighted asymmetry second order, skewed ones first order and visible. The flow gap varies across states; its welfare counterpart hardly does: by the envelope theorem, the consumption-equivalent gap has K -derivative $1 - \beta$ times the panel-(a) gap, whichs consumption-equivalent slope is the panel-(a) lens scaled by $1 - \beta$.

Conditional boundary condition The recursive formulation imposes a stochastic analogue of the transversality condition, necessary in expectation (Kamihigashi, 2003, 2005). Its conditional counterpart is $\lim_{t \rightarrow \infty} \beta^t u'(c_t) K_t = 0$ along each frozen-regime saddle. Holding along every history, it delivers the expectation condition by dominated convergence, capital being bounded and marginal utility finite along the saddle; the converse fails, since the frozen branch has probability zero. Appendix J establishes it from strict concavity rather than from a terminal condition.

Economies without conditional saddle paths A conditional saddle requires a predetermined endogenous state. Without one, the economy jumps each period to the allocation implied by the current exogenous state, so the frozen-A conditional “saddle” collapses to a singleton; I call such an economy *endogenously stateless*.¹² Models with predeter-

adjustment: positive in both regimes, approximately 0.0011 percent of permanent consumption, an optimality check on the computation. The level differences in panel (b) thus primarily reflect the welfare value of regime reversion itself.

¹²The textbook three-equation New Keynesian model under determinacy is endogenously stateless: $(x_t, \pi_t, i_t) = \mathcal{V} s_t$ for exogenous state $s_t := (r_t^n, u_t, v_t)$, so freezing s_t produces no nontrivial dynamics.

mined states (capital, habits, interest-rate smoothing, or distributional states in HANK) are endogenously stateful, and conditional saddle paths exist under mild regularity conditions.

3 State dependence

Responses to the same impulse vary with the business-cycle phase: unemployment dynamics are asymmetric (Neftci, 1984), fiscal multipliers are larger in recessions (Auerbach and Gorodnichenko, 2012), and monetary policy is weaker in recessions (Tenreyro and Thwaites, 2016). I analyze this state dependence through conditional saddle paths. In any stochastic dynamic model admitting such paths, an aggregate variable's response to a shock is a shift across conditional policy graphs. The benchmark is *state independence*: the impact gap between graphs is constant in the endogenous coordinate; departures show up locally as slope differences, the first-order diagnostic.

Lemma 1 (Vertical-separation benchmark).

Let I be an interval over which the regime-specific conditional policy graphs $\mathcal{P}_{A_j}^g = \{(e, g_j(e)) : e \in I\}$, $j \in \{0, 1\}$, are single-valued. Define the impact gap between the two graphs as $\Delta_g(e; A_1, A_0) := g_1(e) - g_0(e)$. Then conditions (i) and (ii) below are equivalent; if g_0, g_1 are differentiable on I , all three are equivalent:

- (i) Δ_g is constant on I ;
- (ii) $\mathcal{P}_{A_1}^g$ is a vertical translation of $\mathcal{P}_{A_0}^g$;
- (iii) the slope gap $g'_1 - g'_0$ vanishes on I .

Proof. The equivalence of (i) and (ii) is definitional. (i) implies (iii) by differentiating the constant gap. (iii) implies (i) by the mean value theorem: $\Delta'_g = g'_1 - g'_0 = 0$ on the interval I makes Δ_g constant. ■

Lemma 1 fixes the benchmark. Proposition 5 makes a substantive claim: the slope gap reflects primitives. Exact state independence is a knife-edge: any differential tilt across conditional policy graphs generates state-dependent shock responses. The degree of state dependence depends on the friction behind the tilt, whether real (Bachmann et al., 2013; Winberry, 2021; Lee, 2026a), financial (Melcangi, 2024), or labor market (Petrosky-Nadeau et al., 2018; Pizzinelli et al., 2020). I measure the tilt locally: each regime's saddle slope at its own conditional steady state, compared across regimes as a ratio. In the log-linearized

frozen-regime RBC dynamics near K_A^{cs} , let $R_A^{cs} := 1 + f'(K_A^{cs}; A) - \delta$ denote the gross return at the conditional steady state, $\lambda_A^s := g'_A(K_A^{cs}) \in (0, 1)$ the slope of the frozen-regime capital map g_A , and $\psi_A := dC/dK|_{\text{saddle}, A}$ the level saddle slope at K_A^{cs} ; each regime's slope is taken at its own K_A^{cs} , so $\mathcal{R}$ below compares two different capital levels.

Proposition 5 (Slope ratio as a structural diagnostic).

The level saddle slope admits the closed form

$$\psi_A = R_A^{cs} - \lambda_A^s, \quad (19)$$

and hence the cross-regime slope ratio is

$$\mathcal{R} := \frac{\psi_B}{\psi_G} = \frac{R_B^{cs} - \lambda_B^s}{R_G^{cs} - \lambda_G^s}. \quad (20)$$

Proof. Log-linearize the frozen-regime budget constraint $K_{t+1} = f(K_t; A) + (1 - \delta)K_t - C_t$ around (K_A^{cs}, C_A^{cs}) :

$$\hat{k}_{t+1} = R_A^{cs} \hat{k}_t - \frac{C_A^{cs}}{K_A^{cs}} \hat{c}_t. \quad (21)$$

Along the frozen-A saddle the deviations satisfy $\hat{c}_t = \psi_A^{\log} \hat{k}_t$ for some log slope $\psi_A^{\log}$. Substituting into (21) yields $\hat{k}_{t+1} = [R_A^{cs} - \frac{C_A^{cs}}{K_A^{cs}} \psi_A^{\log}] \hat{k}_t$, which by definition equals $\lambda_A^s \hat{k}_t$. Solving, $\psi_A^{\log} = (K_A^{cs}/C_A^{cs})(R_A^{cs} - \lambda_A^s)$. Converting to level units via $dC/dK|_{K^{cs}} = (C_A^{cs}/K_A^{cs}) \psi_A^{\log}$ gives (19), and (20) follows by taking the ratio.

Under perfect-foresight expectations, the steady-state Euler condition gives $R_A^{cs} = 1/\beta$ and (19) reduces to $\psi_A = 1/\beta - \lambda_A^s$. Under stochastic expectations over regime transitions, R_A^{cs} is pinned down by $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))(1 + f'(K_A^{cs}; A') - \delta)]$ (the return evaluated at tomorrow's capital K_A^{cs} under the realized regime A') and generally differs from $1/\beta$. Equations (19) and (20) remain valid with R_A^{cs} evaluated at the (stochastic) conditional steady state. ■

Proposition 5 gives $\mathcal{R}$ a structural interpretation: each regime's saddle slope is the gap between the gross return on capital at its conditional steady state and the stable eigenvalue. Once the conditional steady state is located within the stochastic equilibrium, both follow from a local linearization of the frozen-regime dynamics at that point. One reads off (R_A^{cs}, λ_A^s) and applies (20). Local departures from $\mathcal{R} = 1$ indicate state-dependent impact responses. With $\mathcal{R} > 1$ the bad-regime saddle is the steeper, so the gap between the graphs narrows in K and the same shock moves consumption more from a low capital stock; $\mathcal{R} < 1$ reverses the pattern. $\mathcal{R}$ is a model-implied moment identifiable in principle

from state-dependent local projections (Auerbach and Gorodnichenko, 2012; Tenreyro and Thwaites, 2016).

Since $\mathcal{M}(\Phi_0, A)$ is an orbit closure, the slope, curvature, and tangent language of Propositions 5 and 6, of Section 4.1, and of Proposition 10 refers to its local C^1 parametrization near K_A^{cs} , which exists under the regularity conditions of Proposition 10 (Appendix D).

An asymmetric adjustment cost Expanding the capital stock involves installation lags, ramp-up, and congestion, while contracting it largely means letting capital depreciate. If expanding is the more costly margin, the two regimes' policy graphs need not be parallel, and once they are not, the same shock moves the economy by different amounts depending on where it starts. The investment literature points the same way. Bachmann et al. (2013) document stronger aggregate investment responsiveness in expansions, and Winberry (2021) generates procyclical responsiveness through the extensive margin.

The benchmark of Section 2.2 accommodates this with a single change, recalibrated at annual frequency (Appendix H), an adjustment cost whose curvature differs on the two sides of zero net investment: $C(K', K) = \frac{\tilde{\mu}}{2}((K' - K)/K)^2 K$, entering the resource constraint as $C_t + K_{t+1} - (1 - \delta)K_t + C(K_{t+1}, K_t) = A_t K_t^\alpha$, where $\tilde{\mu} = \mu_+$ if $K' - K \geq 0$ and $\tilde{\mu} = \mu_-$ otherwise. Its two summary parameters are the mean curvature $\bar{\mu} := (\mu_+ + \mu_-)/2$ and the asymmetry ratio $r := \mu_+/\mu_-$. Two features of this specification matter. The curvature switches at net accumulation $K' = K$ rather than at gross investment $I = K' - (1 - \delta)K$, so the friction bites when the economy stops adding to its capital stock, not when it stops investing altogether. And the cost is continuously differentiable at that threshold: only the second derivative jumps, so the asymmetry works through curvature and not through a kink in the cost or its slope. I read $\mu_+ > \mu_-$ as a reduced-form expansion bottleneck, capturing installation, ramp-up, or congestion, rather than as a resale wedge.¹³

The threshold and orientation differ from costly reversibility: Abel and Eberly (1996) formalize it as a purchase price above the resale price, and Cooper and Haltiwanger (2006) estimate a mixture of convex, nonconvex, and resale-price costs. Zhang (2005) uses

¹³The calibration sets $\bar{\mu} = 3.05$ and $r = 3$, hence $(\mu_-, \mu_+) = (1.53, 4.57)$. The remaining annual parameters are $\beta = 0.96$, $\delta = 0.10$, and $\alpha = 0.33$; $A \in \{0.98, 1.02\}$ has transition matrix with diagonal entries 0.90. Appendix H details the calibration: the ratio is disciplined by the downside/upside semideviation ratio of annual investment growth (PNFICA, 1967–2019) rather than the raw skewness, which is fragile to the 2009 outlier, and $\mathcal{R}$ is reported across the robustness band $r \in [2, 4]$. That moment bounds r from below at about 2.3 without bounding it from above, so $r = 3$ is a conservative choice inside the identified set rather than a structural estimate of the two curvature parameters. Relative investment volatility is untargeted, since even the frictionless limit sits within one standard error of the narrow series; the level is anchored on the investment– q elasticity.

asymmetric quadratic curvature with $\theta^- > \theta^+$ switching at gross investment $I = 0$, which puts the high curvature on disinvestment rather than on expansion. The specification is a parsimonious setting for studying how an asymmetric real friction tilts policy graphs.

Two comparative statics follow: the first moves the friction pair (μ_-, μ_+) at fixed mean curvature, the second risk aversion (Figure 4). All same-capital comparisons use interior common-support domains. The closed form of Proposition 5 is frictionless, so with adjustment costs I measure the ratio from the solved policies. Within a given specification, define the same- K counterpart $\mathcal{R}(K) = C'_B(K)/C'_G(K)$ over the intersection of the regimes' 5th–95th percentile policy supports, C_A being the single-valued recursive policy since K is the full endogenous state; the marginal- q analogue is $\mathcal{R}_q(K) := q_B^{K'}(K)/q_G^{K'}(K)$, the same construction at $g = q^K$, marginal q defined below. Holding $\bar{\mu} = 3.05$ fixed, the median is 1.265 under the calibrated asymmetry ($r = 3$), 0.976 under symmetric curvature ($r = 1$), and 0.752 under the reversed asymmetry ($r = 1/3$). The friction's orientation therefore determines which regime's saddle is steeper, and reversing it moves $\mathcal{R}$ from above one to below it. Under the calibrated tilt the separations narrow as capital rises: over the sampled range the consumption gap between the saddles falls by 14 percent and the marginal- q gap by 36 percent, so how well capitalized the economy is governs how far a given switch moves it.

The upper row of Figure 4 visualizes a general-equilibrium comparative static on the geometry itself: the saddle pair for $A = B, G$ is the object compared, and preferences are the parameter moved. For this exercise, utility extends from the log benchmark to CRRA, $u(c) = c^{1-\gamma}/(1-\gamma)$, with γ the coefficient of relative risk aversion. Every case keeps $r = 3$, equivalently $(\mu_-, \mu_+) = (1.53, 4.57)$ at $\bar{\mu} = 3.05$, together with the same technology, shock process, and simulated shock path. The left column reports consumption saddles $C_A(K)$; the right reports marginal $q_A^K(K) = 1 + C_{K'} = 1 + \bar{\mu}(K' - K)/K$. The top row varies the CRRA coefficient γ , changing relative risk aversion and the elasticity of intertemporal substitution (EIS), $1/\gamma$, jointly. As γ rises from 1 to 10, the median $\mathcal{R}$ falls from 1.265 to 1.047. The median $\mathcal{R}_q$ falls from 0.468 to 0.321 as both q saddles flatten. But the conditional level gap $\mathbb{E}[q^K | G] - \mathbb{E}[q^K | B]$ rises from 0.0153 to 0.0315, and the simulated unconditional $\text{sd}(q^K)$ rises from 0.00932 to 0.01723. Marginal q therefore becomes less sensitive to capital position within a regime but more volatile because regime switches traverse more widely separated saddles.

The bottom row of Figure 4 turns to Epstein–Zin utility, holding the EIS fixed at one while relative risk aversion rises from 1 to 10. Strikingly, the curves barely move: the

median $\mathcal{R}$ remains approximately 1.265, the median $\mathcal{R}_q$ moves only from 0.4684 to 0.4685, and $\text{sd}(q^K)$ changes only from 0.00932 to 0.00933. The strong CRRA comparative statics come mainly from the accompanying fall in the EIS, not from higher risk aversion, consistent with Tallarini (2000). The anticipation gap of Figure 3 inherits this invariance.

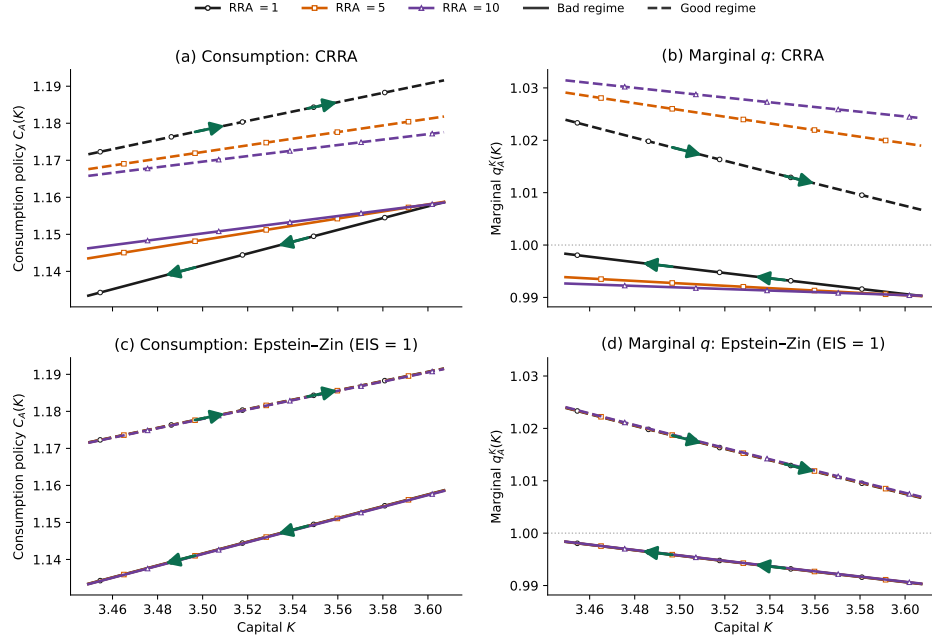


Figure 4: Consumption and marginal- q saddles across preferences

Notes: Panels (a) and (c) plot the consumption policies $C_A(K)$; panels (b) and (d) plot marginal $q_A^K(K)$. The top row varies risk aversion under CRRA, the bottom under Epstein-Zin at unit EIS. Colors and markers code risk aversion; solid and dashed lines the bad and good regimes. All curves are evaluated on the common support of the solved cases. Horizontal lines mark $q^K = 1$; arrows give the direction of capital motion from the sign of $q^K - 1$. With $q_B^K < 1 < q_G^K$ throughout, both conditional steady states lie outside the band: the economy is always in transit. Marginal q displays the assumed adjustment wedge and is not independent calibration evidence. Same-mean symmetric and sign-reversed cases are in Appendix H.

Having used the asymmetric friction to isolate slope-driven state dependence, I return to the frictionless benchmark for a distinct channel: response-shape variation from initial position along the saddle.

Response shape in the frictionless benchmark Although $\mathcal{R} \approx 1$ in the benchmark RBC economy, the *shape* of the response trajectory is highly state dependent. A positive productivity shock displaces the economy across saddles; the trajectory depends on K_0 's position relative to K_G^{cs} . Far below K_G^{cs} , the marginal product of capital is high, tilting the intertemporal tradeoff toward accumulation and producing a late consumption hump (Barro and Sala-i-Martin, 1992); near K_G^{cs} , across-saddle reversion dominates and the peak arrives

earlier. This channel is orthogonal to the slope asymmetry of Proposition 5: slope differences govern *impact* magnitudes, position-dependent convergence speed governs response *shapes*. The shape does *not* require a nonlinear saddle: even in the log-linearized economy the transition superimposes the shock on convergence from K_0 , so its peak timing varies with K_0 , while the isolated shock response does not. The mechanism is a race between the endogenous along-saddle rate λ and the exogenous across-saddle rate ρ , weighted by initial distance from steady state; concavity amplifies the effect but is not its source.

Near the deterministic steady state at the mean of A , distinct from either K_A^{cs} , the Blanchard–Kahn decomposition (Blanchard and Kahn, 1980) on the saddle branch yields

$$\hat{c}_t = \underbrace{\mathcal{P}(\hat{k}_0)}_{\text{along saddle}} \lambda^t + \underbrace{Q}_{\text{across saddle}} \rho^t, \quad (22)$$

where $\hat{c}_t$ and $\hat{k}_0$ are log-deviations from steady state, $\lambda \in (0, 1)$ is the stable eigenvalue of the capital dynamics, and $\rho \in (0, 1)$ is the TFP persistence. The shock size $\varepsilon > 0$ enters both terms. Initial capital enters only the along-saddle coefficient $\mathcal{P}(\hat{k}_0) = \psi_k [\hat{k}_0 + \theta\varepsilon/(\lambda - \rho)]$, where $\psi_k > 0$ is the saddle slope and $\theta > 0$ the capital-TFP elasticity; the across-saddle coefficient $Q = [\psi_a - \psi_k\theta/(\lambda - \rho)]\varepsilon$, with $\psi_a > 0$ the TFP loading of the same policy, is independent of $\hat{k}_0$.¹⁴ The hump arises from the tension between two decaying exponentials: the λ^t term captures along-saddle propagation, the ρ^t term direct TFP reversion, and the peak occurs where they balance. Lower initial capital reduces $\mathcal{P}(\hat{k}_0)$, weakening the slowly decaying along-saddle term, so the balance tips later, exactly the comparative static (23) below.

Proposition 6 (State-dependent peak timing).

Suppose $\lambda > \rho$, $\mathcal{P}(\hat{k}_0) > 0$, and the consumption response (22) admits a peak at $t^* > 0$. Then

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)} < 0, \quad (23)$$

where $\hat{k}_0$ is the log-deviation of the initial capital stock from steady state.

Proof.

See Appendix B. ■

¹⁴This λ (at that steady state) and the regime-specific λ_A^s of Proposition 5 (at the conditional steady state) coincide to first order in the frictionless RBC ($\lambda_B^s = \lambda_G^s = \lambda$, so $\mathcal{R} \approx 1$); frictions generate $\mathcal{R} \neq 1$. Proposition 6 then operates in the $\mathcal{R} \approx 1$ regime, its state dependence driven by initial location $\hat{k}_0$ rather than regime-varying slopes.

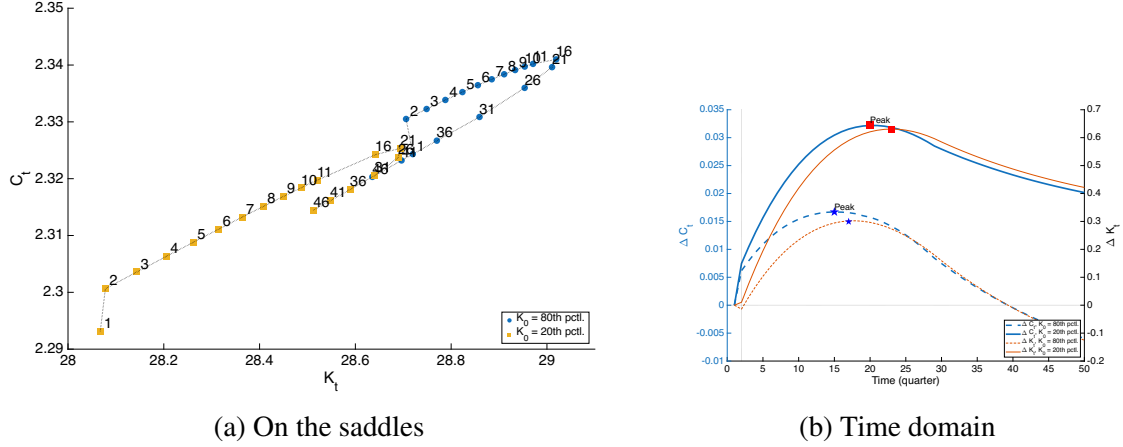


Figure 5: Shape of the post-shock transition: low K_0 vs. high K_0

Notes: Transition paths after a positive productivity shock from K_0 at the 20th and 80th percentiles of the ergodic distribution, each measured from its own pre-shock level. Panel (a): post-shock trajectories in (K, C) space on the conditional saddle paths; numbers indicate quarters since impact. Panel (b): time paths relative to the pre-shock level, ΔC_t on the left axis and ΔK_t on the right; markers indicate the peak quarters. Sample: 50 quarters.

The consumption peak arrives strictly later when initial capital is farther below the steady state. The hypotheses are mild: $\lambda > \rho$ holds under standard quarterly RBC calibrations ($\lambda \approx 0.96$, $\rho \leq 0.95$), and $\mathcal{P}(\hat{k}_0) > 0$ requires only $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$, which the ergodic distribution satisfies with substantial margin. Proposition 6 is local by construction, and Figure 5 shows the mechanism operating globally. Comparing the 20th and 80th percentiles of the ergodic capital distribution, from low K_0 both C and K rise longer before reversion dominates, producing a later and larger consumption peak, while from high K_0 the peak arrives earlier. The same saddle geometry, shock, and preferences apply; only the inherited capital stock differs.

4 Applications

The two applications extend the RBC benchmark of Section 2.2 in complementary directions. The heterogeneous-household economy (Section 4.1) adds idiosyncratic income risk; monotonicity lets aggregate capital index each spell of the frozen orbit, despite its true distributional state. The investment-externality economy (Section 4.2) instead adds investment-generated aggregate knowledge: the second inherited stock turns the family of frozen paths into a *surface*, and the knowledge return private decisions omit identifies the distortion and the optimal policy.

4.1 Heterogeneous households: dimension reduction

In a heterogeneous-household economy, the natural state is the entire cross-sectional wealth distribution, an infinite-dimensional object. Classical aggregation ([Gorman, 1961](#)) reduces dimensionality by restricting preferences, but these restrictions fail under uninsurable idiosyncratic risk and borrowing constraints. Yet practitioners routinely approximate the state by a single variable, aggregate capital. The conditional saddle framework explains why this works without preference restrictions by splitting the claim into two halves of very different difficulty: *within* a frozen-regime spell the distribution evolves along a one-dimensional orbit, so aggregate capital is an *exact* coordinate for it. *Across* histories, the harder half, a single capital rule works only approximately, an error this section bounds and documents directly.

I consider a continuum of ex-ante homogeneous households of unit measure, with recursive problem

$$v(a, z; \Phi, A) = \max_{c, a'} \log(c) + \beta \mathbb{E}v(a', z'; \Phi', A') \quad (24)$$

$$c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z \quad (25)$$

$$a' \geq 0 \quad (26)$$

$$\Phi' = \Gamma_{\text{endo}}(\Phi, A), \quad A' \sim \Gamma_{\text{exo}}(A'|A). \quad (27)$$

The problem differs from the RBC benchmark economy only in (i) uninsurable idiosyncratic labor productivity ([Bewley, 1986](#); [Huggett, 1993](#); [Aiyagari, 1994](#)), a Markov process $z \sim \Gamma_z(z'|z)$; (ii) inclusion of the distribution Φ in the aggregate endogenous state; and (iii) the corresponding law of motion. The RCE is defined as in [Krusell and Smith \(1998\)](#).

The model has two exogenous stochastic processes, idiosyncratic productivity and aggregate TFP, hence two layers of conditional saddle paths, individual and aggregate. I focus on the aggregate path under stochastic TFP, following [Krusell and Smith \(1998\)](#).¹⁵ Consistent with Section 2.2, $K = K(\Phi)$ and $C = C(\Phi, A)$ denote aggregate capital and consumption.

The conditional K -nullcline of Definition 5 extends directly to the heterogeneous-household setting: net investment depends on Φ only through the aggregates (K, C) , so the

¹⁵In the original [Krusell and Smith \(1998\)](#) model exogenous individual labor supply co-moves with aggregate TFP, and the computation follows that specification. Aggregate labor L_A is constant within a regime, so output is $AK^\alpha L_A^{1-\alpha}$ and the frozen-regime analysis is the fixed-labor one with A replaced by the effective productivity $AL_A^{1-\alpha}$. I write A for that product below. With the unemployed supplying 0.25 efficiency units, regime-conditional labor is $L_B = 0.929$ and $L_G = 0.970$, and the effective regime gap is 4.9 percent, 2.4 times the raw TFP gap.

locus remains an object in the aggregate phase plane.

Definition 6 (Conditional K -nullcline in the heterogeneous-household setting).

Fix $A \in \{B, G\}$. The conditional K -nullcline is the zero-net-investment locus in the aggregate (K, C) plane,

$$N(A) := \{(K, C) \in \mathbb{R}_{++} \times \mathbb{R}_{++} : \Delta K(K, C; A) = 0\}, \quad (28)$$

where $\Delta K(K, C; A)$ is the one-period change in aggregate capital implied by the aggregate resource constraint at (K, C) under regime A . As in the representative-agent case, $N(A)$ is distinct from the realized crossings $\{(K_t, C_t) : K_{t+1} = K_t\}$ along any frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$, which under strict K -monotonicity contain no point at all before the limit.

Krusell and Smith (1998) posits an endogenous law of motion tracking aggregate capital K rather than the full distribution Φ : $\log K' = b_0(A) + b_1(A) \log K$. This embeds two distinct assumptions: (i) aggregate capital K suffices to summarize the endogenous aggregate state, and (ii) K follows a log-linear law of motion. Even the existence of a recursive equilibrium on a small endogenous state space is delicate (Kubler and Schmedders, 2003).

Since Krusell and Smith (1998), it has been an open question whether K -based forecasting works so well ($R^2 > 0.9999$) by accident or through a deeper structural property.¹⁶ The within-spell half of (i) is settled exactly: strict monotonicity along a single orbit, established below, makes K index the distributional state path by path, so conditioning on K loses no within-spell information. The across-history half asks why approximately the *same* rule works from every distributional state, not merely along one orbit; I return to it after the within-spell results.

By contrast, (ii) is a different kind of assumption. Even with K as the coordinate, the true law of motion for K is generally non-Markovian, because the omitted distribution still influences next period's capital, and it need not be log-linear, so any parametric rule remains an approximation. This is a question of functional form rather than coordinate choice, and the paper's subject is the latter.

The following theorem is the first step toward exact conditional sufficiency of K . If the economy converges along a one-dimensional equilibrium path and some aggregate variable moves strictly in one direction throughout, its current value identifies exactly where the economy is, since no two distinct distributional states share the same value.

¹⁶Den Haan (2010) shows the forecasting R^2 can substantially overstate dynamic accuracy, because one-step-ahead fit masks the accumulation of errors over long simulated paths; Appendix I applies his stricter multi-period standard, iterating the rule within each spell.

Theorem 1 (Monotone aggregate variable as a coordinate).

Fix (Φ_0, A_0) and $A \in \{B, G\}$. Let $\{\Phi_t\}_{t \geq 0}$ be the frozen-regime continuation $\Phi_{t+1} = \Gamma_{\text{endo}}(\Phi_t, A)$ and let $\Phi^{cs} := \lim_{t \rightarrow \infty} \Phi_t$. Let $e : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ be an aggregate equilibrium variable. Let $t^* \leq \infty$ be the first date at which $\Phi_t = \Phi^{cs}$. Suppose that the scalar sequence $\{e(\Phi_t)\}_{t \geq 0}$ is strictly monotone for $t < t^*$ and converges to $e(\Phi^{cs})$. Then e uniquely indexes states on the conditional saddle:

$$e(\Phi) = e(\Phi') \implies \Phi = \Phi' \quad \forall \Phi, \Phi' \in \mathcal{M}(\Phi_0, A). \quad (29)$$

Proof. Define $\iota_A : \mathbb{N} \cup \{\infty\} \rightarrow \mathcal{M}(\Phi_0, A)$ by $\iota_A(t) := \Phi_t$ for $t \in \mathbb{N}$ and $\iota_A(\infty) := \Phi^{cs}$. By definition of $\mathcal{M}(\Phi_0, A)$ and $\Phi_t \rightarrow \Phi^{cs}$,

$$\mathcal{M}(\Phi_0, A) = \{\Phi_t : t \geq 0\} \cup \{\Phi^{cs}\}, \quad (30)$$

so ι_A is surjective. Take first $t^* = \infty$. Strict monotonicity of $\{e(\Phi_t)\}$ implies $e(\Phi_t) \neq e(\Phi_\tau)$ for $t \neq \tau$, hence $\Phi_t \neq \Phi_\tau$, so ι_A is injective on $\mathbb{N}$; convergence to $e(\Phi^{cs})$ gives $e(\Phi_t) \neq e(\Phi^{cs})$ for all finite t , so ι_A is injective on $\mathbb{N} \cup \{\infty\}$, hence a bijection. If instead $t^* < \infty$, the orbit is the finite set $\{\Phi_0, \dots, \Phi_{t^*}\}$, on which e takes $t^* + 1$ distinct values by strict monotonicity up to t^* , so ι_A restricted to $\{0, \dots, t^*\}$ is again a bijection. For $\Phi = \iota_A(s)$, $\Phi' = \iota_A(s')$, $e(\Phi) = e(\Phi')$ gives $e(\iota_A(s)) = e(\iota_A(s'))$; since $e \circ \iota_A$ is injective, $s = s'$ and $\Phi = \Phi'$. Thus e is injective on $\mathcal{M}(\Phi_0, A)$, and an inverse map $\varphi_{\Phi_0, A}$ exists on $e(\mathcal{M}(\Phi_0, A))$. ■

Consequently, a unique inverse map $\varphi_{\Phi_0, A}$ from $e(\mathcal{M}(\Phi_0, A))$ to $\mathcal{M}(\Phi_0, A)$ satisfies $\varphi_{\Phi_0, A}(e(\Phi)) = \Phi$ on $\mathcal{M}(\Phi_0, A)$, and any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ is a single-valued function of the scalar coordinate e . The following proposition shows that the conditional K -nullclines coincide with those of the representative-agent model.

Proposition 7 (Conditional K -nullclines of Krusell and Smith (1998)).

The conditional K -nullclines of the heterogeneous-household model are identical to those of the representative-household model and invariant over the initial distribution Φ_0 :

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \text{ for } A \in \{B, G\}. \quad (31)$$

Proof.

As in Proposition 3, the stationary condition $\delta K = I$ and the aggregate resource constraint $Y = C + I$ immediately imply (31). ■

As anticipated after Proposition 1, this is the distribution-invariance instance of that common root, and it is what enables dimension reduction. The boundary governing the sign

of capital accumulation, an accounting identity independent of *who* saves, does not move as the distribution evolves along the conditional saddle.

The invariance is specific to the neoclassical production structure with exogenous labor supply. Under separable CRRA preferences with wealth effects on labor, labor supply introduces a distributional channel into the resource constraint; the nullcline then acquires distributional dependence, blocking the analytical route to K -sufficiency. However, monotonicity remains verifiable in those setups: the Appendix G certifies it along the computed frozen-regime continuation, without invoking nullcline invariance.

Proposition 8 (K monotonicity and injectivity).

Fix $A \in \{B, G\}$. Under Assumptions 1 and 2, together with unique K -stationarity along the frozen- A continuation (the no-plateau condition of Section 2.2), aggregate capital converges to K_A^{cs} strictly monotonically on $\mathcal{M}(\Phi_0, A)$. Consequently, K is injective on $\mathcal{M}(\Phi_0, A)$.

Proof.

By Proposition 7, the conditional K -nullcline takes the RBC form, so strict monotonicity follows the proof of Proposition 2 verbatim with $C_A^{Knull}(K) = AK^\alpha - \delta K$. Unique K -stationarity enters locally through the stable-eigenvalue condition and globally as a pathwise hypothesis verifiable by inspecting the simulated frozen-regime continuation. The primitive MPC condition (Appendix E, Lemma E.1) ensures no side-switching but does not rule out non-steady-state flat steps. Since $\{K_t\}$ is strictly monotone and converges to K_A^{cs} , Theorem 1 with $e(\Phi) = K(\Phi)$ yields $K(\Phi) = K(\Phi') \implies \Phi = \Phi'$ for all $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$. ■

Proposition 7 makes the K -nullcline *sign-determining*: the graph $C = C_A^{Knull}(K)$ partitions the (K, C) plane into regions with $\Delta K > 0$ and $\Delta K < 0$ via $\Delta K = C_A^{Knull}(K) - C$. Assumption 2 rules out discrete-time jumps across this partition, so net investment cannot reverse sign along the conditional saddle. Adding unique K -stationarity (the no-plateau condition) to rule out non-steady-state flat steps, capital dynamics are strictly monotone and each frozen-regime spell admits an exact one-dimensional representation indexed by K . Proposition 8 makes $K : \mathcal{M}(\Phi_0, A) \rightarrow \mathbb{R}$ one-to-one, so an inverse map $\varphi_{\Phi_0, A}$ exists with $\Phi = \varphi_{\Phi_0, A}(K(\Phi))$ on $\mathcal{M}(\Phi_0, A)$.¹⁷ Any equilibrium object restricted to $\mathcal{M}(\Phi_0, A)$ then admits an exact single-valued representation as a function of (K, A) .

Panel (a) of Figure 6 plots the computed conditional saddle paths in the (K, C) phase diagram. The model is solved globally using the repeated transition method, and the

¹⁷This is a path-specific notion of sufficiency: for a fixed (Φ_0, A) , K uniquely indexes states on the entire conditional saddle (globally along the transition path).

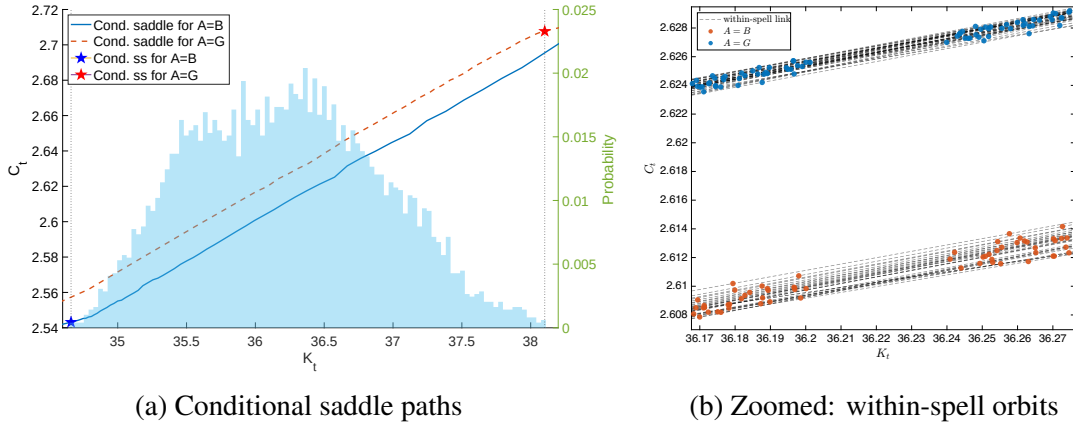


Figure 6: Conditional saddle paths in [Krusell and Smith \(1998\)](#)

Notes: Panel (a): conditional saddle paths for $A = B$ (solid) and $A = G$ (dashed); the background histogram plots the time-series distribution of aggregate capital. Panel (b): simulated ergodic (K_t, C_t) pairs in a narrow window around the median capital stock, $K \in [36.17, 36.28]$ ($A = B$ red, $A = G$ blue); each dashed line fits one spell's points and extends across the window, exposing that spell's local slope.

dynamics under each frozen aggregate state are simulated for 2,500 periods. Each regime's apparent single curve in fact overlays many conditional saddle paths, one per realized spell and its inherited distribution Φ_0 . Although the true endogenous state is Φ rather than K alone, each path is strictly monotone in K , supporting the spell-wise sufficiency result.

Panel (b) zooms into a narrow K -window and uncovers the finer structure: the ergodic band, line-like at the scale of panel (a), is “dusty”. Within each spell the consecutive (K_t, C_t) points move monotonically in K , tracing an exact one-dimensional orbit, while ergodic mixing across spells produces the band observed at any given K . The within-spell links are moreover visibly parallel: the local slope is approximately invariant to the inherited distribution, anticipating its eigenvalue anchor in [Proposition 10](#).

From spell-wise to global sufficiency [Proposition 8](#) is stated spell-by-spell: K uniquely indexes the conditional saddle $\mathcal{M}(\Phi_0, A)$ under a single frozen regime. The equilibrium path concatenates such one-dimensional segments, each exactly indexed by (K, A) within its spell. The handoff between spells, however, is governed by the full distributional state Φ_τ at the switch date, which (K_τ, A_τ) need not determine: different histories can arrive at the same (K_τ, A_τ) carrying different distributions. Spell-wise sufficiency is thus distinct from unconditional global sufficiency, which would require the same K to pin the same Φ regardless of path history. The following proposition bounds this across-spell discrepancy.

Observing the economy at a given K admits a “cloud” of distributions from different prior

histories; three results follow. Theorem 2 supplies the mechanism: economies forget where they started. Proposition 9 averages that over the number of recent regimes two histories share, which the persistence of the exogenous regime process determines. Proposition 10 instead supplies a rate, identifying the local contraction along the saddle with the stable eigenvalue that sets its slope.

The first two rest on a single condition on the frozen-regime maps $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$, which the following definition states and the across-spell analysis maintains; Proposition 10 instead needs local regularity at Φ_A^{cs} (Appendix D), and its eigenvalue gives the on-saddle rate rather than the global λ_c .

Definition 7 (Uniform frozen-regime contraction).

The family $\{\Gamma_A\}_{A \in \{B, G\}}$ is a uniform frozen-regime contraction on $\mathcal{U}$ at rate $\lambda_c \in (0, 1)$ if $\mathcal{U} \subseteq \mathcal{X}$ is closed in the complete metric space $(\mathcal{X}, d)$, forward-invariant under every Γ_A (each Γ_A maps $\mathcal{U}$ into itself), of finite diameter $\text{diam}(\mathcal{U}) := \sup_{\Phi, \Phi' \in \mathcal{U}} d(\Phi, \Phi')$ (the largest distance between any two states in $\mathcal{U}$), and

$$d(\Gamma_A \Phi, \Gamma_A \Phi') \leq \lambda_c d(\Phi, \Phi') \quad \text{for all } \Phi, \Phi' \in \mathcal{U} \text{ and all } A. \quad (32)$$

Theorem 2 (Synchronization under frozen-regime contraction).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7). Then:

- (i) *(Random fixed point: the history-dependent analogue of a contraction's unique fixed point) For every regime history $\omega = (\dots, A_{t-2}, A_{t-1})$, the long-history limit (the state obtained by applying the history's maps from an ever more distant start)*

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} (\Gamma_{A_{t-1}} \circ \Gamma_{A_{t-2}} \circ \dots \circ \Gamma_{A_{t-T}})(\Phi) \quad (33)$$

exists, is independent of $\Phi \in \mathcal{U}$, and is the unique state in $\mathcal{U}$ consistent with ω ; any trajectory initialized in $\mathcal{U}$ at date $t - T$ satisfies $d(\Phi_t, \Phi^(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U})$.*

- (ii) *(Suffix synchronization) If two histories ω, ω' agree over their most recent T periods, then*

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (34)$$

Proof.

See Appendix C. ■

It is worth noting that the limit (33) in part (i) runs backward. Fix the recent regime sequence and start the economy ever earlier: because every map contracts, the starting

state washes out and the recent history alone pins down where the economy is now. This is a random-fixed-point construction in the spirit of [Schenk-Hoppé and Schmalfuß \(2001\)](#), and the suffix bound (34) is its consequence for two states in the same- K cloud, averaged over the common-suffix length in (36). For the stationary symmetric two-state chain ($\pi_{BB} = \pi_{GG} = \rho_A$), let $q := \rho_A^2 + (1 - \rho_A)^2$ denote the probability that two independent histories agreeing at a date also agree one period earlier.¹⁸

Proposition 9 (Across-spell approximation bound).

Let $\{\Gamma_A\}$ be a uniform frozen-regime contraction on $\mathcal{U}$ at rate λ_c (Definition 7), and let $\Phi, \Phi' \in \mathcal{U}$ be the long-history limits of two regime histories that agree over their most recent T periods. Then

$$d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (35)$$

If, moreover, the two histories are drawn independently from the stationary symmetric chain and share the current regime, then $\Pr(T = k) = (1 - q)q^k$ for $k \geq 0$ and

$$\mathbb{E}[d(\Phi, \Phi')] \leq \frac{1 - q}{1 - q\lambda_c} \text{diam}(\mathcal{U}). \quad (36)$$

Proof.

See Appendix C. ■

Each period of shared regime history contracts the distance between the two long-history states by the factor λ_c , starting from the worst case $\text{diam}(\mathcal{U})$; the pathwise bound (35) compounds this over the T shared periods. The expectation bound (36) averages over T , which is geometric under the stationary chain: the expected gap is small when spells are persistent (large q) and the contraction is strong (small λ_c). The count T begins at the previous date, because the shared current regime indexes the cloud but has not yet acted on Φ . The diameter $\text{diam}(\mathcal{U})$ is finite under the borrowing constraint $a' \geq 0$ and compact idiosyncratic support. The right-hand sides involve only the primitive set $\mathcal{U}$, not the same- K cloud itself, so the proposition is a genuine bound on the cloud rather than a contraction statement about it. The bound (36) vanishes as $\rho_A \rightarrow 1$, since long spells make recent suffixes long. It does not vanish as $\lambda_c \rightarrow 0$: a pair sharing no regime history is bounded only by $\text{diam}(\mathcal{U})$, and that happens with probability $1 - q$.

Ergodic interpretation Theorem 2(i) extends Proposition 9 to the ergodic set. Every state observed along a long simulated path is, up to a geometrically small error, the long-history

¹⁸Stepping back one period, two independent histories still agree if both persist (ρ_A^2) or both switch ($(1 - \rho_A)^2$); for asymmetric chains, q is the backward agreement probability of the time-reversed chain.

limit of its own realized history. Every pair in the same- K cloud is therefore a pair of long-history limits, inheriting the pathwise bound (35) with T the realized common-suffix length. This applies at every date, deep within a spell or just after a switch, since the argument runs over histories rather than spells.

The expectation bound (36), by contrast, is computed under the idealized model of two independent stationary histories, and two caveats attach: within-path pairs are dependent, and conditioning on a common K is a selection. It is therefore a benchmark for the typical scale of the cloud rather than a bound on its realized diameter.

Theorem 2 leaves λ_c unspecified. The next proposition anchors it along the saddle near the conditional steady state: the stable eigenvalue of the linearized frozen-regime dynamics, the same object governing the local saddle slope (Proposition 5). This holds under the regularity of infinite-dimensional linearization: smoothness, a spectral gap, metric-norm agreement, and a stable manifold carrying the orbit tail, the last maintained rather than implied by the others. Expressing the rate in terms of aggregate capital requires a fifth condition: the aggregate capital functional $\Phi \mapsto K(\Phi)$ is continuously differentiable near Φ_A^{cs} , with derivative nonzero along the stable direction. Appendix D states the five conditions precisely and explains each briefly.

Proposition 10 (Local contraction rate on the conditional saddle).

Near the conditional steady state, the frozen-regime dynamics on the conditional saddle contract distances at the geometric rate $|\lambda_A^s|$, where $\lambda_A^s \in (0, 1)$ is the stable eigenvalue of the linearized frozen-regime dynamics; in terms of aggregate capital,

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|). \quad (37)$$

Proof.

See Appendix D. ■

The full statement on the distributional state space, its regularity conditions, and the proof are in Appendix D. Directions orthogonal to the saddle may contract at different rates, so the global rate λ_c in Theorem 2 could exceed $|\lambda_A^s|$. The proposition delivers the primitive anchor, the same eigenvalue that drives the slope-ratio diagnostic (Proposition 5) and disciplines the local contraction in Theorem 2 and Proposition 9.

Assumed uniform contraction The hypothesis that $\Gamma_{\text{endo}}(\cdot, A)$ contracts uniformly on $\mathcal{U}$ is not innocuous: the map embeds general-equilibrium price feedback, and establishing

contraction-type properties for such operators is precisely the difficulty confronted by global existence and uniqueness analyses (Proehl, 2025b). The eigenvalue anchor of Proposition 10 is local; globally the hypothesis is maintained rather than verified: Proposition 9 requires λ_c to be bounded above, while the diagnostic below measures only the contraction the computed map exhibits, so even $\lambda_c < 1$ is assumed. This does not conflict with the saddle structure, since Γ_{endo} is the equilibrium law of motion after the forward-looking variable has been solved out, leaving the unstable direction of the (K, C) system outside the distributional dynamics.

A sample-based diagnostic for λ_c The global uniform λ_c in Theorem 2 bounds contraction across a cloud of distributional states at the same K reached from different histories, which need not lie on a single conditional saddle. The diagnostic below can show the map contracts no faster than some observed rate; it cannot certify that it contracts at least that fast for every pair of states, which is what (35) needs. Since the across-spell bound uses λ_c as an *upper* bound on the per-step rate, a certified verification of Proposition 9 requires a certified upper bound on λ_c . Such a bound would come via either a Lipschitz constant for Γ_{endo} on $\mathcal{U}$ (model-specific) or a statistical confidence bound on sample estimates (standard but tedious in function spaces). This is beyond the scope of the paper and left to future work. Instead, I propose a sample-based *diagnostic* that measures the one-step contraction actually exhibited by the computed map; Santos and Peralta-Alva (2005) give conditions under which moments of simulated approximate equilibria converge to their exact counterparts.

The diagnostic draws N pairs of distributional states at common K and regime from the simulated equilibrium, applies the frozen-regime map to both members of each pair, and reports $\hat{\lambda}_c^N := \max_i r_i$ over the one-step ratios r_i . The ratios are computed in total variation on the joint distribution over assets and employment. The map is non-expansive in that metric by construction, so $r_i \leq 1$ holds automatically and the diagnostic measures how far below one the observed rates fall. A contraction rate depends on the metric, not only the topology, so this does not bound the λ_c of Definition 7, and no such inequality is claimed. Substituting it in (36) illustrates what that bound delivers at an observed rate rather than bounding it.

Under the Krusell and Smith (1998) calibration, quarterly regime persistence is $\rho_A = \pi_{AA} = 0.875$, so spells last eight quarters on average and $q = \rho_A^2 + (1 - \rho_A)^2 = 0.781$. Over some 45,000 same- K pairs the diagnostic gives a median one-step ratio of 0.982 and a maximum $\hat{\lambda}_c^N \approx 0.9999$. The sample thus shows almost no contraction margin. At that

rate the expectation bound (36) leaves a factor of 0.9997 multiplying $\text{diam}(\mathcal{U})$, so it is uninformative. Proposition 10 anticipates the gap: the deterministic benchmark saddle contracts at 0.965, but directions off the saddle contract far more slowly, and the uniform rate is set by the slowest of them. A direct check is more informative. Comparing the globally solved K' at each spell boundary with the prediction from tracking only (K, A) gives an average discrepancy of 2.5×10^{-4} in logs, 1.2 percent of one standard deviation of $\ln K$. This is a mean forecast error in K' , not a worst-case distance between distributions, so it does not verify the bound above; it documents separately that the across-spell error is small here.

State reduction, functional form, and block recursivity The Krusell and Smith (1998) approach is often labeled near-rational (Akerlof and Yellen, 1985), but this conflates two sources of approximation the conditional-saddle structure separates. Within an orbit, conditioning on K is exact, and only the functional form of the law of motion approximates. This within-orbit exactness connects the conditional sufficiency of K to *block recursive equilibrium* (BRE), where aggregate prices and quantities follow from a low-dimensional state (Menzio and Shi, 2010, 2011; Schaal, 2017). In directed-search environments block recursivity holds by construction; here it emerges spell-wise from the saddle geometry, with prices, quantities, and Φ recoverable from K along a given conditional saddle. Across histories, by contrast, a residual state-reduction error remains, small in this calibration, which the framework identifies rather than assumes away; additional forecasting moments can reduce it, with gain bounded by the across-spell discrepancy. This is where K -sufficiency is weaker than exact aggregation, in which (K, A) alone determines the aggregates regardless of history: recovery from K also requires knowing which conditional saddle the economy is on, and the regime history that generated the spell selects it.

Verifying monotonicity in practice Theorem 1 and Proposition 8 justify the repeated transition method (Lee, 2026a), which uses K as the conditional sufficient statistic in heterogeneous-agent models, with K indexing the equilibrium state without pinning down the functional form of the law of motion. The framework also yields a practical tool: a cheap check of the underlying monotonicity hypothesis (Assumption 2 with unique K -stationarity), run on the computed equilibrium itself: a method that assumes scalar-state sufficiency solves first, then certifies the assumption on its own output or flags its failure, rather than leaving it maintained. The check comes in two forms. The first is an *order-*

preservation check (Appendix G), which delivers within-spell K -monotonicity from the computed frozen-regime map, using that map directly where it is scalar and proceeding pathwise per realized spell otherwise: a strictly increasing map with a unique stable fixed point converges monotonically, and in the [Krusell and Smith \(1998\)](#) economy all 84 realized frozen-regime spells are strictly monotone in K , with rank correlations of K_{t+1} on K_t of 0.9999 and 1.0000 consistent with, but not establishing, order preservation. The second works from primitives (Appendix E): bounds on household MPCs and price feedback exclude side-switching under rational expectations, yielding exact within-spell K -sufficiency under log utility and Cobb–Douglas production. This second route is still verified along a computed continuation rather than from primitives alone, with two conditions maintained explicitly, a pathwise finite-difference bound on aggregate consumption and the residual no-plateau condition. The logic extends to bonds and GHH labor, and to multiple predetermined states where a monotone coordinate exists (Section 4.2), its boundary the lumpy-investment economy ([Khan and Thomas, 2008](#); [Bachmann et al., 2013](#)) where nullcline invariance fails and monotonicity becomes a testable diagnostic rather than a theorem (Appendix I).

4.2 An investment externality: the conditional saddle surface

This second application adds a second predetermined stock: investment also builds aggregate knowledge H , which augments productivity but is not priced by firms. This combines the cumulative-investment learning of [Arrow \(1962\)](#), the nonappropriable knowledge of [Romer \(1986\)](#), and the two-stock propagation of [Cooper and Johri \(2002\)](#).¹⁹ With two predetermined stocks, the frozen orbits implied by the equilibrium dynamics form a surface. Solving the planner’s problem over the same two stocks traces a second surface, which rests at higher capital and knowledge. Under homogeneity both surfaces become cones, and the capital-knowledge ratio recovers a one-dimensional conditional saddle path that characterizes the full equilibrium dynamics.

An endogenous-productivity augmentation Relative to Section 2.2, the only addition is a second stock. Aggregate investment builds a stock of knowledge H that augments productivity, so effective productivity and the two accumulation equations are

$$\tilde{A}_t := A_t H_t^\eta, \quad K_{t+1} = (1 - \delta)K_t + I_t, \quad H_{t+1} = (1 - \delta_H)H_t + \gamma I_t^\phi, \quad (38)$$

¹⁹[Benkard \(2000\)](#) documents organizational forgetting, which motivates the knowledge depreciation $\delta_H > 0$. Appendix K gives the supporting evidence, the planner derivation, the surface validation, and the replication code.

where $g(I) := \gamma I^\phi$ denotes the knowledge that aggregate investment generates. The exponent $\phi \in (0, 1]$ is the elasticity of new knowledge to investment, so $\phi < 1$ makes learning subject to diminishing returns and $\phi = 1$ is the constant-returns benchmark; η is the elasticity of productivity to the knowledge stock, the strength of the spillover. Because knowledge enters production only through $\tilde{A}_t$, output is $Y_t = \tilde{A}_t K_t^\alpha = A_t K_t^\alpha H_t^\eta$, the resource constraint $C_t + I_t = Y_t$ is unchanged, and the price schedules (12) carry over with A_t replaced by $\tilde{A}_t$.

The representative household of Section 2.2 carries over unchanged and remains atomistic in the sense of price taking: firms take the knowledge stock as given and households its aggregate law of motion, so no private decision internalizes the effect of investment on H_{t+1} . Only A_t is exogenous, so freezing the regime still leaves H_t , and hence $\tilde{A}_t$, evolving. The private Euler equation is therefore the one from Section 2.2 evaluated at $\tilde{A}_t$, and it excludes the knowledge return:

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{1}{C_{t+1}} \{1 + r(K_{t+1}, \tilde{A}_{t+1})\} \right] = \beta \mathbb{E}_t \left[\frac{1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} H_{t+1}^\eta}{C_{t+1}} \right]. \quad (39)$$

The stationary calibration satisfies $\alpha + \eta\phi < 1$; the homogeneous limit below sets $(\eta, \phi) = (1 - \alpha, 1)$. The quarterly calibration is standard, and the three knowledge parameters are illustrative rather than estimated, with the spillover deliberately small, so the geometry rather than the size of the correction carries the argument.²⁰

The planner and the credit Because investment jointly produces next-period capital and knowledge, the planner carries *two* co-states, $p = V_K$ and $q = V_H$, alongside $\lambda = 1/C$:

$$\begin{aligned} \lambda_t &= \beta \mathbb{E}_t[p_{t+1}] + g'(I_t) \beta \mathbb{E}_t[q_{t+1}], & p_t &= \lambda_t \alpha \frac{Y_t}{K_t} + (1 - \delta) \beta \mathbb{E}_t[p_{t+1}], \\ q_t &= \lambda_t \eta \frac{Y_t}{H_t} + (1 - \delta_H) \beta \mathbb{E}_t[q_{t+1}]. \end{aligned} \quad (40)$$

In the standard model, substituting the investment condition $\lambda_t = \beta \mathbb{E}_t[p_{t+1}]$ into the capital envelope collapses the shadow value of capital to $p_t = \lambda_t(1 - \delta + \alpha Y_t/K_t)$. That step fails here: the investment condition in (40) carries the extra knowledge term, so the shadow value of installed capital lies below that benchmark by $(1 - \delta)g'(I_t)\beta \mathbb{E}_t[q_{t+1}]$, and the knowledge return enters as a *flow* on gross investment, not as a capitalized addition to the price of capital. A Pigouvian investment credit that prices the spillover,

$$\tau_t = \frac{g'(I_t) \beta \mathbb{E}_t[q_{t+1}]}{\lambda_t}, \quad (41)$$

²⁰The standard block is $\beta = 0.99$, $\delta = 0.025$, $\alpha = 0.36$, and $A \in \{0.99, 1.01\}$ with persistence $\pi_{AA} = 0.95$. Knowledge adds $(\delta_H, \phi, \eta) = (0.10, 0.7, 0.01)$, with γ normalizing the deterministic competitive H^* to one. Appendix K reports the solution and its validation.

restores efficiency. Because I_t , q_{t+1} , and λ_t depend on the inherited stocks, the credit is itself a surface over (K, H) , the policy counterpart of the allocation surfaces.

At the calibration the deterministic planner steady state lies 3.9 percent above the competitive one in capital and 2.7 percent in knowledge, with $\bar{\tau} = 0.0242$. The steady-state credit depends on (η, ϕ) only through $m := \eta\phi$: the unit-investment condition, the first line of (40), splits into a capital piece and a knowledge piece, and the credit is the second. In closed form, $\bar{\tau} = bm/(a + bm)$, where $a = \alpha/[1 - \beta(1 - \delta)]$ and $b = \delta_H/\{\delta[1 - \beta(1 - \delta_H)]\}$ discount the two returns. Since $\bar{\tau}$ is small, doubling the spillover very nearly doubles the credit: its elasticity in m is $1 - \bar{\tau} \approx 0.98$. The magnitudes reported below therefore scale with the assumed spillover, while the qualitative results do not.

Implemented as a gross-investment credit financed by a lump-sum levy $\mathcal{T}_t = \tau_t I_t$, taken as given by the household, it lowers the effective price of gross individual investment $i_t := k_{t+1} - (1 - \delta)k_t$ to $1 - \tau_t$ with a balanced budget each period. The credited household budget and Euler equation are

$$\begin{aligned} c_t + (1 - \tau_t)i_t &= \frac{\alpha Y_t}{K_t} k_t + w_t - \mathcal{T}_t, \\ (1 - \tau_t)\lambda_t &= \beta \mathbb{E}_t \left[\lambda_{t+1} \left\{ \frac{\alpha Y_{t+1}}{K_{t+1}} + (1 - \tau_{t+1})(1 - \delta) \right\} \right], \end{aligned} \quad (42)$$

where undepreciated capital is valued at the net-of-credit replacement price. Substituting the planner conditions (40) into (42) shows that the credit (41) reproduces the planner's allocation exactly. The same holds numerically: feeding the credit through the credited Euler equation along the computed planner path leaves a mean residual of 2.8×10^{-6} , no larger than the solution's own error (Appendix K).

The conditional saddle surface In the computed basin each pair of initial stocks launches a frozen-regime orbit $\mathcal{M}((K_0, H_0), A)$, agents still pricing switches; the closure of their union, $\mathcal{S}_A$, is the consumption-policy surface over (K, H) . Figure 7(a) traces it by drawing the frozen orbits themselves. The deterministic-equation benchmark has two stable roots against one unstable, so the surface is a genuine saddle.²¹ The conditional steady states (37.07, 0.983) and (38.92, 1.017) bracket the cloud's capital range, $K \in [37.10, 38.91]$, because capital converges monotonically within every spell. Knowledge behaves differently: built by gross investment, which runs above its steady-state level while capital is still rising,

²¹Appendix K reports the counts, eigenvalue magnitudes, and frozen-map classification: the frozen-map Jacobian is stable at B and resolution-ambiguous at G : differencing the interpolated policy surface along a near-neutral direction leaves the slow root's magnitude spanning 0.98–1.04. The dedicated frozen paths still converge to the conditional steady states, at decay rate 0.96 under G and 0.94 under B , establishing attraction without identifying the second local eigenvalue.

it overshoots and then recedes. It peaks inside 181 of the 642 spells, and the cloud reaches slightly beyond the two points in knowledge, $H \in [0.979, 1.021]$. The two stable roots separate the stocks and set these speeds: the fast root 0.903 is almost purely a knowledge direction at the depreciation rate $1 - \delta_H = 0.90$, against the capital-dominated slow root 0.965. Knowledge therefore settles quickly toward $H = \gamma I^\phi / \delta_H$, whose steady-state form $H = \gamma(\delta K)^\phi / \delta_H$ is common to both regimes, and the economy crawls along it in capital. The ergodic set is a thin ribbon about that locus, departing by 0.8 percent on average, and the planner inherits the structure. Since the mean regime spell is also twenty quarters, the economy usually switches regime before capital adjustment is complete: the conditional steady states are attractors it approaches but rarely reaches.

The planner's surface and where it rests The planner faces the same two stocks, so its frozen dynamics trace a second surface over the same base, itself a saddle with two stable roots against three, counted in the five-variable system (K, H, C, q, p) : the static consumption condition contributes an infinite root, and eliminating C leaves two stable against two finite unstable (Appendix K). The two surfaces are not usefully compared point by point, because the economies they describe do not occupy the same states: the externality moves the whole configuration rather than tilting one surface against a fixed other. What is well defined is where each surface rests. From the same frozen-regime solves, the planner's conditional steady states are (38.54, 1.010) and (40.43, 1.045), against the equilibrium's (37.07, 0.983) and (38.92, 1.017): the planner rests at higher capital and higher knowledge in both regimes. Since each economy clusters around its own pair, the two recurrent sets barely meet. The planner's ergodic capital range [38.59, 40.43] overlaps the equilibrium's [37.10, 38.91] only over [38.59, 38.91], and 12.7 percent of planner dates fall inside the equilibrium's range. At any common state the planner consumes 1.3 to 2.9 percent less, saving more; operating at higher capital, its ergodic mean consumption is nonetheless 0.6 percent higher, at an essentially equal log standard deviation.

The tax credit surface Figure 7(b) draws the optimal tax credit (41) on the same base as the allocation surfaces. It is state-dependent and countercyclical, averaging 2.45 percent in the bad regime against 2.39 percent in the good, because low investment raises the marginal knowledge return $g'(I_t)$. The same force makes it rise with capital, while diminishing returns to knowledge make it fall with H . A constant credit matches only the deterministic planner steady state and need not implement the planner in the stochastic economy. Here the

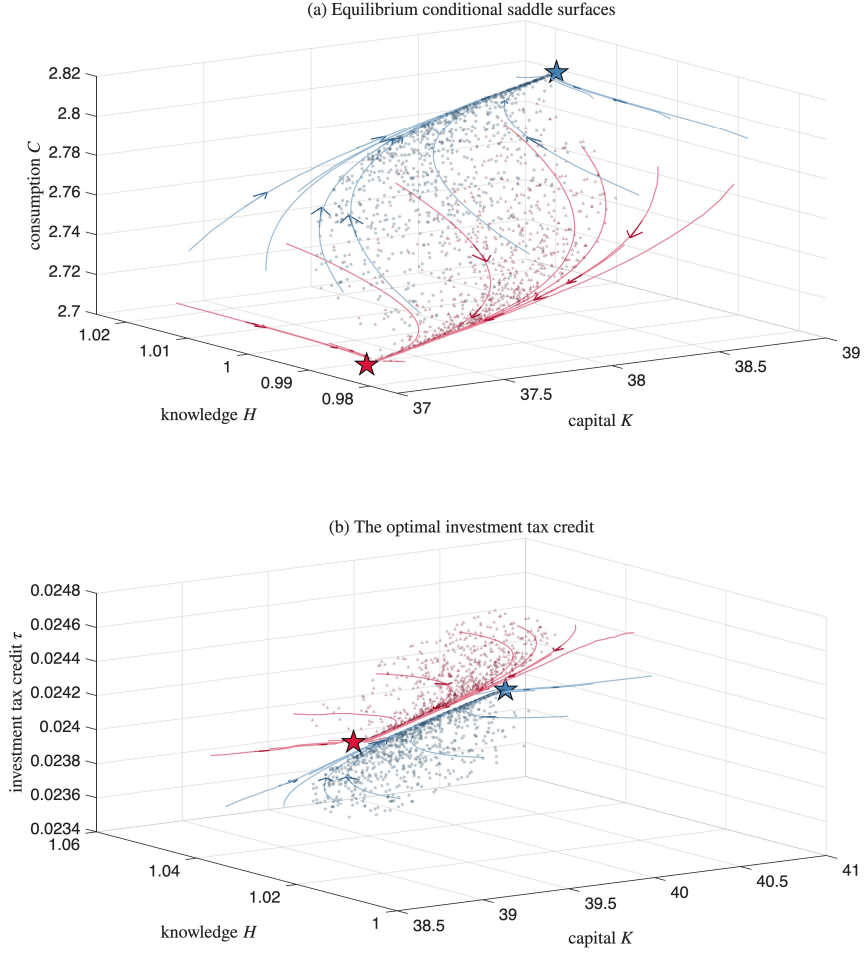


Figure 7: The conditional saddle surface and the optimal investment tax credit.

Notes: Panel (a): the good-regime (steel blue) and bad-regime (crimson) equilibrium conditional saddle surfaces, each drawn as a family of frozen orbits converging into the conditional steady states (stars; Appendix K); arrows mark the direction of motion. Panel (b): the optimal investment tax credit (41) as a surface over the same base, drawn in the same format from frozen planner orbits, with stars at the planner's conditional steady states.

two nearly coincide even though the surface is genuinely tilted, with elasticities of $+0.6$ in capital and -0.5 in knowledge; the ergodic set spans about four percent in each stock, so τ_t stays within $[0.0237, 0.0247]$. From a common anchor under identical shocks, $\bar{\tau} = 0.0242$ therefore tracks the planner within 0.1 percent over eight hundred quarters. Matching the planner's steady state is thus a weaker requirement than implementing its policy along transitions: the constant credit is anchored at a deterministic benchmark, not even at the

conditional steady states, while the surface is exactly right everywhere.

Boundary of the construction The surface exists because the composite returns satisfy $\alpha + \eta\phi < 1$, which gives two stable roots for the two predetermined stocks (Appendix K). At equality, a unit root appears, and above the boundary only one stable direction remains, so generic two-stock convergence fails. This is a loss of saddle convergence, not sunspot indeterminacy: indeterminacy would require more stable roots than predetermined states, whereas this economy has fewer. In the exact homogeneous case, $\phi = 1$ and $\eta = 1 - \alpha$, the frozen-regime system is homogeneous of degree one, the surface becomes a cone, and the ratio K/H is exactly sufficient for the normalized dynamics. This is dimension reduction by symmetry rather than by the monotonicity of Section 4.1. However, the two coincide here: the ratio is strictly monotone along all realized spells in the simulation (642 spells), so it serves as a monotone coordinate as well (Appendix K).

5 Concluding remarks

Conditional saddle paths, equilibrium trajectories under frozen exogenous regimes, extend the geometric analyses of deterministic phase diagrams to stochastic environments. In particular, this paper highlights three distinguished properties of the conditional saddle paths. Slopes capture state dependence: differential tilt reflects asymmetric shock responses, and distance along the saddle shapes the response even at nearly equal slopes. Monotonicity delivers dimension reduction: a strictly monotone aggregate exactly indexes the state within each frozen-regime spell, the structural reason scalar-state methods work. With a second predetermined stock, the path becomes a surface and the planner's rests above it, the omitted knowledge return marking the distortion and the policy that closes it: an investment credit, itself a surface over the two stocks, state-dependent and countercyclical.

More broadly, questions that have required separate numerical results, such as when a scalar state suffices, how responses depend on position in the state space, and what anticipation is worth, become legible in a single geometric representation. This framework opens a new pathway for stochastic equilibrium analysis: state-dependent local projections can estimate its slope and timing predictions from the data, disciplining the primitives behind them, and its policy surfaces extend to stabilization rules that condition on the aggregate state. Thus, with this geometric device, the stochastic equilibrium dynamics can be tightened to reflect observation, and policy more precisely implemented and evaluated.

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